Standard Procedures for Regulation of the Willamette Basin Projects



US Army Corps of Engineers Portland District

This document outlines the standard procedures for regulation of the Willamette projects. It is meant to be used as a quick reference and does not replace the Water Control Manual.

Date	Revision Number	Description
03 Dec 12	R0	Reformatted the previous SOP dated 11 November 2011 and added reference to Special Operations.
22 Nov 13	R1	Updated Figure 25 for more accurate turbine flow calculations and removed Scullion's home phone number.
04 June 14	R2	Updated Figure 23b, Figure 25, Section 10 links and Section 11 spreadsheet references.
30 Jan 15	R3	Table 1 storage for LOP, FAL, DET. Link ref updated from Water_Control\Willamette_Basin\Regulation to Reservoir_Regulation\Willamette_Basin\Regulator_Information. Updated contacts. Verified all links. Updated file names to current year.
02 Oct 15	R4	Updated Sections 2.1, 3.3, 4.1 and 4.2; updated contacts; added drought year operational notes; and added project outlet considerations for Green Peter.
19 Apr 17	R5	Updated contacts, links, and Figure 25. Reservoir and outlet work stats to include lowest operable outlet information.
5 Jun 17	R6	Add information concerning total dissolved gas and add new temperature targets below Big Cliff Dam (Table 11).
15 Jul 19	R7	Overall update
30 Jan 20	R8	Update language regarding outflow of HCR and CGR during flood season

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1 Overview

This document outlines the standard procedures for regulation of the Willamette projects. It describes day-to-day regulation activities and procedures as well as roles and responsibilities. It is meant to be used as a quick reference and does not replace the water control manuals or the Dam Safety Emergency Action Plans (EAPs). This is a "living" document and should be updated as procedures change.

Congress originally authorized the construction of the U.S. Army Corps of Engineers' Willamette Project for the purposes of power, recreation, fish and wildlife, and irrigation. These projects also aid to improve downstream water. ***It is important to remember that the number one seasonal priority for the Corps is flood damage reduction*.** All other purposes are secondary with the order of importance varying depending on circumstances.

The Willamette Basin contains 13 projects. There are three types of projects in the Willamette River system: base load projects, non-power projects, and power projects.

- The **base load** projects are typically on-line all the time and are not used for peaking. *Hills Creek* and *Cougar* are base load projects.
- **Non-power** projects have no generating capability. *Blue River, Fern Ridge, Cottage Grove, Dorena,* and *Fall Creek* are non-power projects.
- The **power projects** generate power for peaking purposes and have a re-regulation project immediately downstream and are as follows: **Lookout Point***/Dexter,* **Green Peter***/Foster,* and **Detroit***/Big Cliff*. The re-regulation project's purpose is to smooth out flows downstream of the power projects. They will run steady, much like the baseload projects, with their pool level varying on a daily cycle.

The Willamette Projects, shown in Figures 1 through 13 below, control about 27% of the watershed area above Portland. They were designed to "effectively" control the 1861 flood. Figure 14 shows the location of the projects within the Willamette Basin.

Figure 1: Aerial view of Big Cliff dam and reservoir.



Project: Big Cliff Dam

Stream: North Santiam River

Subbasin: North Santiam Basin

Type: Re-Regulating for Detroit Dam

Figure 2: Aerial view of Blue River dam and reservoir.



Project: Blue River Dam

Stream: Blue River

Subbasin: McKenzie Basin

Type: Multistorage; Non-Power

Figure 3: Aerial view of Cottage Grove dam and reservoir.



Project: Cottage Grove Dam

Stream: Coast Fork Willamette River

Subbasin: Coast Fork Basin

Type: Multistorage; Non-Power

Figure 4: Aerial view of Cougar dam and reservoir.



Project: Cougar Dam

Stream: South Fork McKenzie River

Subbasin: McKenzie Basin

Type: Multistorage; Baseload

Figure 5: Aerial view of Detroit dam and reservoir.



Project: Detroit Dam

Stream: North Santiam River

Subbasin: North Santiam Basin

Type: Multistorage; Power Peaking

Figure 6: Aerial view of Dexter dam and reservoir.



Project: Dexter Dam

Stream: Middle Fork Willamette River

Subbasin: Middle Fork Willamette Basin

Type: Re-Regulating for Lookout Point Dam

Figure 7: Aerial view of Dorena dam and reservoir.



Project: Dorena Dam

Stream: Row River

Subbasin: Coast Fork Basin

Type: Multistorage; Non-Power

Figure 8: Aerial view of Fall Creek dam and reservoir.



Project: Fall Creek Dam

Stream: Fall Creek

Subbasin: Middle Fork Willamette Basin

Type: Multistorage; Non-Power

Figure 9: Aerial view of Fern Ridge dam and reservoir.



Project: Fern Ridge Dam

Stream: Long Tom River

Subbasin: Coast Fork Basin

Type: Multistorage; Non-Power

Figure 10: Aerial view of Foster dam and reservoir.



Project: Foster Dam

Stream: South Santiam River

Subbasin: South Santiam Basin

Type: Multistorage; Re-Regulating

Figure 11: Aerial view of Green Peter dam and reservoir.



Project: Green Peter Dam

Stream: Middle Santiam River

Subbasin: South Santiam Basin

Type: Multistorage; Power Peaking

Figure 12: Aerial view of Hills Creek dam and reservoir.



Project: Hills Creek Dam

Stream: Middle Fork Willamette River

Subbasin: Middle Fork Willamette Basin

Type: Multistorage; Baseload

Figure 13: Aerial view of Lookout Point dam and reservoir.



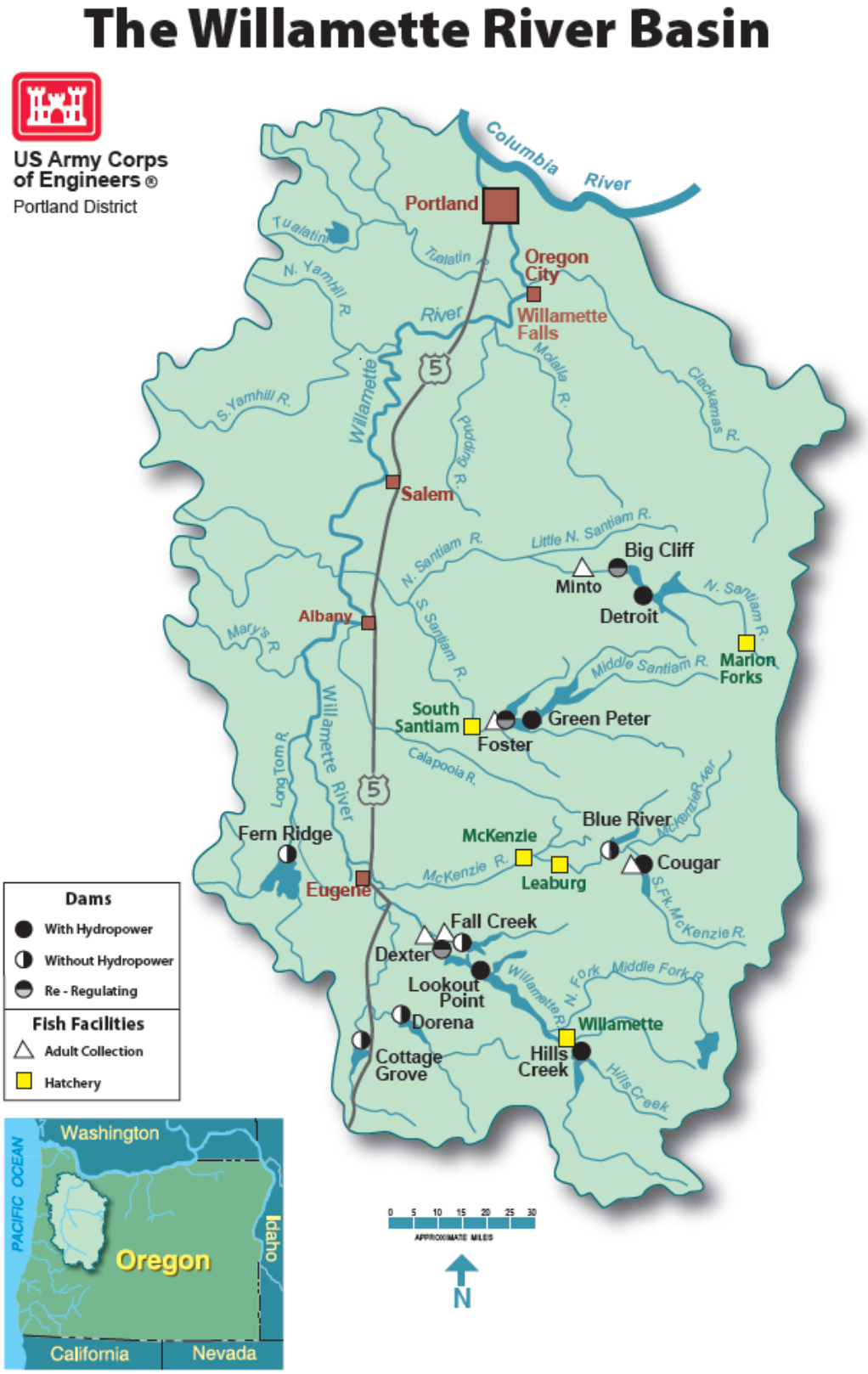
Project: Lookout Point Dam

Stream: Middle Fork Willamette River

Subbasin: Middle Fork Willamette Basin

Type: Multistorage; Power Peaking

Figure 14: Map of the Willamette Basin

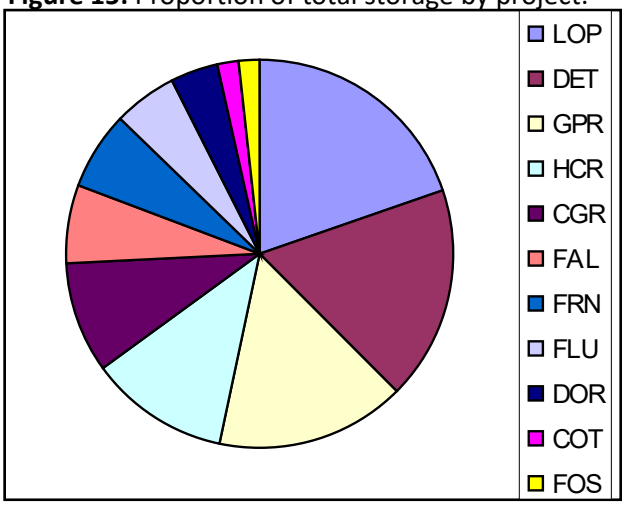


The three power peaking projects (Lookout Point, Detroit, and Green Peter) account for more than half of the annual conservation storage in the basin. Table 1 and Figure 15 below show the proportion of storage (to full pool) available in each of the 11 storage projects.

Table 1: Storage quantity by project.

Project	Storage (Acre-Feet)	% of Total Storage
Lookout Point (LOP)	337,400	20%
Detroit (DET)	300,200	18%
Green Peter (GPR)	268,250	16%
Hills Creek (HLC)	200,200	12%
Cougar (CGR)	155,180	9%
Fall Creek (FAL)	113,700	7%
Fern Ridge (FRN)	109,620	6%
Blue River (BLU)	85,550	5%
Dorena (DOR)	70,510	4%
Cottage Grove (COT)	29,790	2%
Foster (FOS)	29,710	2%
Total	1.7 million acre-feet	

Figure 15: Proportion of total storage by project.



2 Standard Regulation Procedures

2.1 Daily Activities

Daily activities include the steps outlined below.

1. Update the daily spreadsheet called "CWMS Daily Update.xlsm." At least once a week, the "Summerblock" worksheet should also be copied to the "Daily project graphs.xlsm" spreadsheet. See Section 11.3 for more information on this spreadsheet tool.
2. Check all available data first thing in the morning, prior to any briefing or running the forecast. Evaluate reservoir, stream gage and weather data and make a general assessment of conditions for focus areas. Also check generation reports to obtain latest generation rates. See Section 10 for a description of available data sources.
3. Determine project inflow and current outflow. Compute inflow if necessary using the change in storage and outflow for at least the latest 6 hour time period. There is a convenient inflow calculator worksheet in the "Will_Schedules.xlsm" workbook.
4. Determine storage/elevation in relation to rule curve. The goal is to remain at or below the flood control rule curve unless regulating an event. There will be occasions when the reservoir fills above the flood control rule curve, but this should be temporary while regulating the event. It is permissible for the reservoir elevation to be below the rule curve (e.g. when following a modified fill schedule, drafting the reservoir, slowly ramping down outflow after an event), due to lack of inflow, or during summer use of conservation storage for downstream augmentation; but should not go below minimum conservation pool unless generating per BPA request.
5. Run the Northwest River Forecast Center (NWRFC) Community Hydrologic Prediction System (CHPS) model after they call to tell us it's ready for our use. This includes looking back at observed conditions for recent days and inputting planned regulation for the short-term forecast period. The NWRFC usually calls by 9:00 to tell us that the model is ready. Project outflow input into the model should be based on the forecasted flows as shown in the NWRFC CHPS. This may not be the same as what is actually scheduled because observed flows may be different than what is forecast. We can coordinate with NWRFC to run the model more than once daily and also on the weekends if needed. See Section 11.1 for information on running the CHPS model.
6. Evaluate the forecast in CHPS and/or using spreadsheet tools. If necessary, discuss planned regulation with the regulation team to decide on a plan of action. The rule of thumb is not to change releases until conditions warrant (e.g. increased flows at control points or rising inflows), unless good judgment deems it necessary.
7. Compute the project outflow needed to meet operational goals (e.g. provide flood damage reduction, draft back to the rule curve after an event, fill the reservoir at the desired rate, or meet downstream flow goals during conservation season). Check rate of change limitations (ramp rates) for project outflow or downstream locations, minimum/maximum release requirements for normal and special operations, and any tailwater restrictions (see Table 12).
8. Talk to Greg Taylor (OD-V) or Chris Walker (OD-S) to review possible fisheries concerns associated with an operation.

9. Rerun CHPS model with new regulations and see if forecast flows at downstream gages are within limits, reservoirs are within limits and project objectives are being met. The CHPS model should be run and approved before 10am so NWRFC can put out their official forecasts.
10. Calculate generation hours available and provide the hours to BPA for load shaping. Provide BPA with the number of unit hours available at the power peaking projects (Detroit, Green Peter and Lookout Point) so they can prepare the generation schedule. Provide BPA the estimated megawatts (MW) scheduled at the baseload projects (Cougar and Hills Creek). Call the 3-Shift Hydro Desk if the schedule is for the next day (or later) and call the Real-Time Hydro Desk if a change is being made to the schedule for the current day. BPA will call back with the generation schedule. BPA has requested that schedules be called in before 11am.
11. Prepare schedules and send to the projects via the Columbia Basin Teletype (CBT) Messenger. Project release schedules should be completed and sent out as soon as possible in the morning. Typically schedules are sent out on Mondays and Thursdays, but can be sent out as often as needed when operations change. After schedules have been sent via the CBT, prepare an email that sends the schedules to the wider distribution list. See Section 2.2 for information on preparing and sending schedules.
12. Participate in daily briefings if scheduled. Regularly scheduled briefing calls are at 1:00 PM on Tuesdays and Thursdays. The briefing call on Tuesdays is with BPA and Bureau of Reclamation while the briefing call on Thursdays is with National Weather Service. There may be additional briefings with the Project, the Division RCC office, and/or Readiness Branch, especially during times of high flow.

2.2 Regulation Schedules

Schedules contain the instructions to the operators for the immediate and or indefinite future. This may include lake releases, water temperature goals, downstream temperature targets, specific instructions for operations, and the immediate and general intent of the described operation. Typically schedules are sent out twice a week on Mondays and Thursdays; however, it is prudent to send schedules more often if needed to confirm operational goals and operational criteria. The Monday schedule will typically contain schedule information for Tuesday through Thursday and the Thursday schedule will contain information for Friday through Monday. When precipitation is forecast, it is often beneficial to send schedules for only the following day to reduce the need to revise schedules.

When creating a schedule, staffing and project characteristics should be considered. Considerations for scheduling include the following:

- Lookout Point Control Room operates Lookout Point, Dexter, Hills Creek, Fall Creek, Cottage Grove, Dorena, Fern Ridge, Cougar and Blue River.
- Foster Control Room operates Foster and remotely operates Green Peter.
- Detroit Control Room operates Detroit and remotely operates Big Cliff.
- Lookout Point Control Room is staffed 24 hours with two 12-hour shifts. Shift changes are at 04:00 and 16:00 hours. Only one operator is generally on per shift and they can be extremely busy.

- Foster Control Room is staffed 24 hours with two 12-hour shifts. Shift changes are at 05:00 and 17:00 hours.
- Detroit Control Room is staffed 24 hours with two 12 hour shifts. Shift changes are at 05:00 and 17:00 hours.
- Lookout Point operators have requested that no spillway changes be scheduled after dark for safety reasons.
- Detroit spillway gate changes are made remotely from the control room so spillway gate changes can be made at any time. Changes to Big Cliff generation and spill are done remotely from the control room and may also be made at any time.
- Use the power unit curves in the "Turbine Calculator.xlsm" to obtain generation rates.
- When possible, notify the operator of impending changes to the Cougar schedule in the morning so the operator can notify Eugene Water and Electric Board (EWEB) Leaburg Dam of upcoming changes.
- The schedules are free-form text, but they must have the appropriate text on the second line of the schedule to ensure proper delivery. Each schedule should also have CENWP/RCC below the regulator's name. Example schedules are shown in Figures 16-18 and stored in a folder under the CENWP-EC-HR Outlook mailbox.
- Non-power schedule should be sent to the "NO" distribution list, baseload schedules should be sent to the "BS" distribution list, power schedules should be sent to the "PW" distribution list, and weekend duty schedule should be sent to the "PD" distribution list on the CBT Messenger.

Sample Willamette non-power schedule.

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
APW R 071519 1021 NO LOP NPD NPB NPP NPC
WILLAMETTE NON-POWER PROJECTS

SCHEDULE FOR: TUE - THU,  16 JUL  -  18 JUL 2019

PROJECT    : APROX TOTAL RELEASE          EFFECTIVE
-----------  --------------------  ------------------
FALL CREEK  :  RELEASE =     647 CFS      15 JUL @ 0900 HRS
MON: REDUCE RO FLOW ~0.75 FT OPEN OR 200 CFS MAX REDUCTION
RO FLOW RELEASE AUGMENTS MAINSTEM FLOW AT ALBANY
CURRENT ELEVATION IS 816 FT
************************************************************
COTT GROVE  :  RELEASE =      51 CFS      15 JUL @ 0900 HRS
TARGET FLOW RANGE OF 50-55 CFS
CURRENT ELEVATION IS 781.3 FT
************************************************************
DORENA     :  RELEASE =     107 CFS      15 JUL @ 0900 HRS
TARGET FLOW RANGE OF 100-105 CFS
CURRENT ELEVATION IS 819.6 FT
************************************************************
BLUE RIVER  :  RELEASE =      51 CFS      15 JUL @ 0900 HRS
TARGET FLOW RANGE OF 50-55 CFS
CURRENT ELEVATION IS 1311.4 FT
************************************************************
FERN RIDGE  :  RELEASE =      50 CFS      15 JUL @ 0900 HRS
MAINTAIN MINIMUM FLOW OF 30 CFS AT MONROE
CURRENT ELEVATION IS 372.6 FT
************************************************************
AFTER HOURS CONTACT NUMBERS:
                CELL PHONE
1. MARY KAREN SCULLION   (503) 927-4642
2. KEVIN MCALLISTER      (503) 318-3524
************************************************************
MARY KAREN SCULLION
CENPW/RCC
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

Sample Willamette baseload schedule.

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
APW R 071519 1024 BS LOP DET FOS BPA NPD NPB NPP NPC
WILLAMETTE BASE LOAD PROJECTS
------------------------------------------------------------
HILLS CREEK:
SCHEDULE FOR: TUE - THU,  16 JUL  -  18 JUL 2019
 APROX GEN   TURBINE   SPILL   TOTAL   EFFECTIVE TIME
 ----------- --------- ------- ------- ----------------
  9      490      0     490   15 JUL @ 0900 HRS
CURRENT ELEVATION IS 1521.6 FT
X
------------------------------------------------------------
COUGAR:
SCHEDULE FOR: TUE - THU,  16 JUL  -  18 JUL 2019
 APROX GEN   TURBINE   SPILL   TOTAL   EFFECTIVE TIME
 ----------- --------- ------- ------- ----------------
  9      310      0     310   15 JUL @ 0900 HRS
CURRENT ELEVATION IS 1595.7 FT
X
ADJUST THE TEMPERATURE TOWER WEIRS AS FOLLOWS:
WEIRS 1,4,7 & 2,5,8 ARE AT ELEV 1562
WEIRS 3,6,9 ~10FT FROM WATER SURFACE
**************************************************
AFTER HOURS CONTACT NUMBERS:
                 CELL PHONE
1. MARY KAREN SCULLION   (503) 927-4642
2. KEVIN MCALLISTER      (503) 318-3524
**************************************************
MARY KAREN SCULLION
CENPW/RCC
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

Sample Willamette power schedule.

```
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
APW R 061019 1057 PW LOP FOS DET NPP NPD NPB BPA

WILLAMETTE POWER AND RE-REG PROJECTS:
---------------------------------------
X
X
LOP - DEX SCHEDULE FOR TUE - THU,  11 JUN - 13 JUN 2019
X
       POWER PROJECT    *     REREG PROJECT
   HOURS  #UNTS SPILL TURB  *  OUTFLOW  GEN.  ELEV.
      00=              *            693.3
  0 -  8 = 0    0    0 *    3600   0.0  690.9
  8 -  9 = 1    0 2630 *    3600   0.0  690.8
  9 - 17 = 2    0 5260 *    3600   0.0  692.0
 17 - 22 = 3    0 7890 *    3600   0.0  693.9
 22 - 24 = 0    0    0 *    3600   0.0  693.3
TOT UNIT HRS=32    RLS=3506. *RLS= 3600    LOCAL=  80
X
DEX U1 CONTINUES TO BE FOS
CONTINUE MAINTSTEM FLOW AUGMENTATION
X
GPR - FOS SCHEDULE FOR TUE - THU,  11 JUN - 13 JUN 2019
X
       POWER PROJECT    *     REREG PROJECT
   HOURS  #UNTS SPILL TURB  *  OUTFLOW  GEN.  ELEV.
      00=              *            636.8
  0 -  6 = 0    0   50 *    1130   2.6  636.4
  6 -  7 = 1    0 1850 *    1130   2.6  636.5
  7 -  9 = 1    0 1850 *    1130   8.0  636.6
  9 - 18 = 0    0   50 *    1130   8.0  636.1
 18 - 19 = 2    0 3700 *    1130   8.0  636.3
 19 - 22 = 2    0 3700 *    1130   2.7  636.9
 22 - 24 = 0    0   50 *    1130   2.7  636.8
TOT UNIT HRS=11    RLS=883.3 *RLS= 1130    LOCAL= 250
X
FOS U2 OOS (MAINTENANCE), ERTS 6/13
MAINTAIN FOS BETWEEN 635-637 FT
FOS NIGHTTIME FISH PASSAGE SPILL CONTINUES UNTIL JUNE 15
POC: FENTON KHAN ((509) 551-7711)
PRIORITIZE SPILL BETWEEN 1700-0700 HOURS, WITH MIN
TURBINE FLOW FOR STATION SERVICE
FOS MIN IS CURRENTLY 1100 CFS @SSFO
X
DET - BCL SCHEDULE FOR TUE - THU,  11 JUN - 13 JUN 2019
X
       POWER PROJECT    *     REREG PROJECT
   HOURS  #UNTS SPILL TURB  *  OUTFLOW  GEN.  ELEV.
      00=              *           1200.7
  0 -  6 = 0  1000    0 *    1300   7.6 1200.0
  6 -  9 = 50    0 1920 *    1300   7.7 1201.5
  9 - 16 = 0    0    0 *    1300   7.2 1195.4
 16 - 19 = 50    0 1920 *    1300   7.4 1197.1
 19 - 22 = 50  1000 1920 *    1300   7.7 1200.9
 22 - 24 = 0  1000    0 *    1300   7.7 1200.7
TOT UNIT MW=450    RLS=1178. *RLS= 1300    LOCAL= 120
X
MAINTAIN APPROX. 37/63 SPILL/GEN RATIO AS SCHEDULED
FOR TEMP. CONTROL OPERATIONS AT DETROIT
MIN BCL RELEASE IS 1200 CFS @BCLO
```

2.2.1 Willamette Scheduling MS Excel Workbook

The Willamette scheduling workbook (Will Schedules.xlsm) is used to prepare the non-power, baseload, and power schedules. The workbook includes sheets for each type of schedule.

Non-power and base load schedules require few calculations. The spreadsheets for those projects do little more than create the first line of the schedule in the predetermined format and insert the dates based on the user specified number of days to schedule. These two spreadsheets show recent flows [pulled from the Corps Water Management System (CWMS) database], plant outages (pulled from the BPA Outage Report), and indicate if the first scheduled flows exceed ramp rates or are below minimum flows. If more than one line is to be used in a schedule (e.g. flows are being changed throughout the schedule period), it must all be inserted and edited manually. When complete, these schedules can be copied and pasted into the CBT Messenger. Additional notes, such as when a schedule is "revised", may be added directly in the CBT Messenger rather than stored in the scheduling workbook. Occasionally schedule changes are minor and the Regulator may prefer to copy a schedule previously sent and edit it in the CBT Messenger rather than use the scheduling workbook.

When creating power schedules, the power project worksheet (in Will Schedules 2015.xlsm) helps significantly. When scheduling generation at the power project, BPA specifies unit schedules in the form "0 for 6, 1 for 3, 0 for 8, 2 for 3, 0 for 4" This means "run 0 units for 6 hours, 1 unit for 3 hours, 0 units for 8 hours, 2 units for 3 hours and 0 units for 4 hours." The schedule should total 24 hours in a day. These numbers are then entered on the "Power" worksheet along with starting reservoir elevations, local flows, project flows and the flow per unit at the current head. A macro is then run to format this data into a schedule on the "Text" worksheet sheet. Notes and on-call contact information can be added to the schedule on the "Text" worksheet.

If you need to revise a schedule place a note under the first line of text with the date and time of the old schedule you are revising, the project that has the revision, and the reason for the revision.

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
APW R 041817 1600 PW LOP FOS DET NPP NPD NPB BPA
REVISE APW R 041817 1501 PW FOR LOOKOUT POINT OUTAGE ON APRIL 19
```

Figure 19 below shows an example of the "Power" worksheet for Lookout Point/Dexter. Only the shaded cells should be modified by the user; other cells should not be changed or deleted.

Example power schedule worksheet for Lookout Point/Dexter.

	---------- LOOKOUT POINT----------						---------- DEXTER ----------						
			recent:	3411			recent:	2707		693.0			1.00 =MWff
THURS	#U	hrs	spill	Qpwr	local	spill	Qpwr	kaf	end El.	Head	MW	Qmax	
1			0	2500	50	0	2500	25367	693.0			15	max
4/20/2017	2	0	6	0	0	50	0	2550	24127	691.7	54	10	
24		3	3	0	7500	50	0	2550	25367	693.0	55	10	
		0	9	0	0	50	0	2550	23508	691.1	53	10	
		3	5	0	7500	50	0	2550	25574	693.2	55	10	
		0	1	0	0	50	0	2550	25367	693.0	55	10	
				0	0	50	0	2550					
				0	0	50	0	2550					
				0	0	50	0	2550					
				0	0	50	0	2550					
				0	0	50	0	2550					
				0	0	50	0	2550					
	0	24	24	0	2500	50	0	2550	25367	693.0		0	

In this example, the flow per unit at the current head (Qpwr) is 2,500 cfs. The left two columns contain the load shaping information provided by BPA. In this example, 24 unit hours were scheduled over 24 hours resulting in an average outflow of 2,550 cfs from Lookout Point. Dexter information is entered to the right of the dividing line. In this example, the starting elevation is 693 ft, the preferred starting and ending elevation for every schedule, the generator is scheduled to run at 10 MW, the local flow is 50 cfs and the release from the turbine at Dexter is 2,500 cfs. This results in an ending Dexter pool of 693.0 ft. For the re-regulating project, the schedule also shows the ending storage and corresponding elevation, daily average local inflow, spill, outflow, and MWs for the day.

In the example above, only five of the allowed 11 lines of changes are used. Showing the additional lines of locals and Dexter outflow has no impact on the schedule. Eleven lines of changes per day is nearly always more than enough to schedule the operation. For each power/reregulation project pairs (Lookout Point/Dexter, Green Peter/Foster, and Detroit/Big Cliff), the block shown in Figure 19 above is repeated five times, one below the other. These blocks allow up to five different schedules to go out on the CBT for each project. The duration of each of the schedule blocks can be specified for any number of days.

Once the macro is run, the schedule in Figure 19 is transformed into text format as shown in Figure 20. Previous schedules in text format are stored in columns to the right of the current schedule. This preserves past schedules to allow the regulator to copy previous notes, points of contact, etc. (The sheet stops functioning when the past schedules exceed 256 columns, the Excel limit. If this happens, delete the right-most columns.)

Example power schedule in text format for Lookout Point.

```
LOP - DEX SCHEDULE FOR THU - FRI,  20 APR - 21 APR 2017
X
           POWER PROJECT      *      REREG PROJECT
    HOURS  #UNTS SPILL TURB   *   OUTFLOW  GEN.  ELEV.
        00=                   *                  693.0
   0 -   6 =  0     0     0   *     2550    9.9  691.7
   6 -   9 =  3     0  7500   *     2550   10.1  693.0
   9 -  18 =  0     0     0   *     2550    9.7  691.1
  18 -  23 =  3     0  7500   *     2550   10.1  693.2
  23 -  24 =  0     0     0   *     2550   10.1  693.0
TOT UNIT HRS=24    RLS= 2500 *RLS= 2550    LOCAL=  50
```

The scheduling workbook also contains other helpful information and functionality. This includes:

- Observed Unit Status is listed so the regulator is reminded how many units are available today, but the BPA Outage Schedule should be checked to verify how many units are available tomorrow.
- The number of unit hours to schedule to maintain the same starting and ending elevation at the re-regulating dam.
- Outflow from the re-regulating project for the last hour of data retreived.
- The efficiency of the re-regulating generators can be input to make calculated and observed MWs consistent.
- At Dexter, when transitioning above or below 1500 cfs notify the local CWMS programmers (currently Art, Ruth, or Ari).
- Schedules can start from zero (current day) to any number of days in the future.
- Schedules can be specified to apply to any number of days.
- The number of lines of comments for each schedule can be specified on the “Power” sheet, or “add rows” and “delete rows” buttons can be used on the “Text” sheet. (These buttons insert cells in column A, and therefore do not delete lines from the saved schedules in the other columns.)
- If zero is listed as the number of days for which the schedule applies, it will not be included in the schedule. This is often used to see how the schedule for the current day is going so as to show the correct starting elevation for the following day’s schedule.
- When daylight savings stops or starts, 23 or 25-hour days are scheduled. The cell showing the total number of hours will become red, but this can be ignored.
- Use of conditional formatting to when Dexter and Big Cliff are scheduled above or below their pool ranges. This may occur when large blocks of generation or large periods of

time with no generation are scheduled, which will cause over-filling or over-draining the re-regulation pools, respectively. Do not schedule projects beyond their normal limits.

- In general all of the Power Projects are sent together in one schedule (PW), the non-power projects are sent together in one schedule (NO) and the baseload projects are sent in one schedule (BS). The acronyms PW, NO and BS are the group names to send the schedules to in the CBT messenger (right hand block at the bottom of the CBT messenger screen).
- When revising a schedule always indicate the schedule that you are revising.

```
APW R 041917 1012 PW LOP FOS DET NPP NPD NPB BPA
REVISE APW R 041917 1012 PW FOR DETROIT
WILLAMETTE POWER AND RE-REG PROJECTS:
```

- Re-send the other schedules with the revised schedule (e.g. resend LOP and GPR if you're revising DET) so that they are all together on the CBT and on the web.

2.2.2 CBT Messenger

- In late summer 1998, RCC switched from flow schedules sent via teletypes to flow schedules sent via the world wide web. Despite being electronic, the process of sending a schedule via the CBT Web Messenger is still known as "sending a teletype." The link to the CBT Web Messenger is: https://cbt2.usace.army.mil/cbtcgi/id1
- The username is g0cbtnwp [that is "G" "Zero"]. The password is cb-CB=56.

2.2.3 Actual Energy Regulation [AER]

The Corps is a member of the PNCA (Pacific Northwest Coordination Agreement) along with BPA, Portland General Electric, and other hydroelectric producers to make sure all power needs are met in the Pacific Northwest. The PNCA hired the Northwest Power Pool (NWPP) to conduct on-going studies to maintain fairness and reliability in the hydropower system. The NPP runs a study that keeps track of "accounting" of group members' rights to power and water, while making sure everybody meets the system load. Twice a month the PNCA members are asked to submit data for the Actual Energy Regulation (AER) for their projects. The AER is necessary for the NPP to determine the regulation of the projects, including the Willamette projects, and the water that will be sent downstream to the next party.

The letter template is located as a worksheet called "pwrpool-AER" inside the "CWMS Daily Update.xlsm" workbook. This is described in more detail in Section 11.3.1. The first data submittal for the AER for the month is done within the first 7 days of the month and the second one is done within the last 10 days of the month. NWPP.Org sends out a reminder that the data submittal is due and identifies the forecast period. There are 14 periods – one each month and two for April and August. From March through July, the forecast goes through July 31st. From August through September, the forecast goes through September 30th. From September to March, the forecast goes through the next two periods. Median values were from DOE-BP-4352 "2010 Modified Streamflow 1928-2008." The

Willamette AER information is sent to the Division Hydrologic Engineering and Power Branch (CENWD-PDW-HP) in the format shown in Figure 21.

Example Willamette AER submittal.

ATTN: Chisholm, Thomas A CIV USARMY CENWD (US) <Tom.A.Chisholm@usace.army.mil>;Tilton, Jeffrey P CIV USARMY CENWD (US) <Jeffrey.P.Tilton@usace.army.mil>; Stanfill, Michael D CIV USARMY CENWD (US) <Michael.D.Stanfill@usace.army.mil>; DLL-CENWP-RCC <DLL-CENWP-RCC@usace.army.mil>

3-Jul-19

Updated Willamette system end of month draft levels are summarized in the following table. Values are in ksfd of draft from maximum conservation pool.

	JUNE 30	JUL 31	AUG 15	AUG 31	SEPT 30
DETROIT	12.72	6.25	10.77	16.65	43.11
FOSTER	0.03	0.00	0.00	0.00	6.46
GREEN PETER	47.56	58.77	65.66	73.75	100.23
COUGAR	46.40	46.33	47.29	48.59	55.43
HILLS CREEK	22.98	24.77	30.24	35.94	46.96
LOOKOUT POINT	80.18	105.87	112.96	122.70	138.71
LOST CREEK	33.72	48.01	62.02	76.02	89.11

	1 - 30 June		1 - 31 July		1 - 15 August		16 - 31 August		1 - 30 September	
	INFLOW	% MEDIAN	INFLOW	% MEDIAN	INFLOW	% MEDIAN	INFLOW	% MEDIAN	INFLOW	% MEDIAN
DETROIT	0.89	0.51	0.93	1.00	0.69	1.00	0.65	1.00	0.67	1.00
FOSTER	0.43	0.34	0.43	1.00	0.23	1.00	0.20	1.00	0.23	1.00
GREEN PETER	0.11	0.16	0.26	1.00	0.14	1.00	0.13	1.00	0.15	1.00
COUGAR	0.38	0.54	0.33	1.00	0.25	1.00	0.24	1.00	0.23	1.00
HILLS CREEK	0.65	0.68	0.44	1.00	0.34	1.00	0.32	1.00	0.31	1.00
LOOKOUT POINT	1.34	0.55	1.07	1.00	0.82	1.00	0.78	1.00	0.76	1.00
LOST CREEK	1.72	0.78	1.26	1.00	1.08	1.00	1.01	1.00	1.00	1.00

2.2.4 "No Load" Requests

BPA, as the regional power marketing agency for hydropower generated at Corps Projects, must manage generation with real time power load requirements. Scheduling generation adds significant complexity to day-to-day operations and some risk to flow operations, in particular minimum flows and ramping rates. Each of the power projects in the Willamette system has different ability to meet short-term BPA scheduling requests. There are, therefore, preferred operational choices when scheduling hydropower generation on a short-term or real-time basis. This scheduling is the planned schedules done in coordination with the Bonneville Power Administration (BPA) "**Real Time**," and "**3-Shift**" schedulers. Requests from the **BPA dispatchers** will be handled differently since they are typically due to a sudden outage or emergency.

The BPA 3-shift schedulers typically request generation availability for the near future. NWP Water Management takes these requests and incorporates generation into project schedules which are sent out via "CBT." BPA real-time schedulers may call on shorter notice as a result of fluctuations in demand and a resulting need to adjust current generation schedules. NWP has only limited flexibility to meet these requests, depending on which project. Green Peter and Hills Creek projects typically have the most operational flexibility. Detroit and Lookout Point have some limited flexibility to manage short term changes in generation. The remaining projects, Cougar, Big Cliff, Dexter, and Foster offer very little to no flexibility. Requests to modify generation at these projects should be considered carefully.

A separate document (provided on pages 24-28) describe operational limitations that may dictate or influence the scheduling of No Load Flexibility (when generation capability exceeds demand) for Portland District hydropower projects in the Willamette and Rogue Basins. The document also describes the preferred operational choices when the reservoir elevations are above and below the spillway crest. Operational flexibility, environmental implementation, and BiOp priorities are subject to change.

NWP Water Management, in cooperation with other Corps offices, has the authority to schedule, or not schedule, generation based on other projects needs and priorities. Denying generation schedule requests should be considered carefully.

No-Load, No Generation Flexibility for NWP non-Mainstem Columbia Hydropower Projects

The purpose of the document is to provide a short description of the preferred operational choices when scheduling hydropower generation on a short-term or real-time basis. The scheduling referred to in this document are the planned schedules done in coordination with the Bonneville Power Administration (BPA) "Real Time," and "3-Shift" schedulers. Requests from the BPA dispatchers will be handled differently since they are typically due to a sudden outage or emergency.

The BPA 3-shift schedulers typically request generation availability for the near future. NWP Water Management takes these requests and incorporates generation into project schedules which are sent out via "CBT." BPA real-time schedulers may call on shorter notice as a result of fluctuations in demand and a resulting need to adjust current generation schedules. NWP has only limited flexibility to meet these requests, depending on which project. Green Peter and Hills Creek projects typically have the most operational flexibility.

This document describes operational limitations that may dictate or influence the scheduling of No Load Flexibility (when generation capability exceeds demand) for Portland District hydropower projects in the Willamette and Rogue Basins. The document also describes the preferred operational choices when the reservoir elevations are above and below the spillway crest. Operational flexibility, environmental implementation, and Biological Opinion priorities are subject to change. This document will be revisited on an annual basis.

1. Projects with the **most flexibility** include Green Peter and Hills Creek. The preferred operational choices for these two projects are described below.

    - **Green Peter**
      The project may change between regulating outlet (RO) flow and generation at any time; however, pre-scheduled transitions between outlets are preferred. Pre-scheduled outflow will determine how many unit hours are available each day.

      Fisheries considerations are as follows:
        - Currently no spring Chinook salmon or winter steelhead inhabits Green Peter Reservoir or the Middle Santiam below Green Peter Dam.
        - No Fisheries or Water Quality operations are planned for 2017.

    - **Hills Creek**
      The project may change between RO flow and generation at any time; however, pre-scheduled transitions between outlets are preferred.

      Caveats to this operation include:
        - SCADA must be operational.
        - The unit may be run at either Speed No Load (which causes the units to run in the lower cavitation zone) or the butterfly valve(s) will be closed. Stopping the units without closing the butterfly valves causes the units to creep, which causes more wear on the bearing.

- To re-energize a unit, staff has to be on-site to manually water up the unit prior to re-start. Therefore, to re-start a unit it is preferred that this occurs during normal staffing hours of 0800-1700 Monday through Thursday.
  - One way to avoid manual start-up at Hills Creek is to install equipment to allow remote control of the water-up valves such as what is in place at Cougar. When the butterfly valves were replaced at Hills Creek the fill valves were motorized but not connected to SCADA.

Fisheries considerations are as follows:

- Spring Chinook salmon are present in Hills Creek Reservoir and below the dam, but no spawning or incubation occurs below the dam.
- Ramp rates should be met during the spring and early summer to avoid stranding outmigrating juvenile salmon leaving Hills Creek Reservoir and to avoid impacting popular trout angling areas below the dam.
- No Fisheries or Water Quality operations are planned for 2017.

2. Projects with **limited flexibility** include Detroit and Lookout Point. The preferred operational choices for these two projects are described below.

- **Detroit**

When the reservoir is below spillway crest, elevation 1541 ft, (typically October 15 through April 7), the project may change between RO flow and generation at any time; however, pre-scheduled transitions between outlets are preferred. Pre-scheduled outflow will determine how many unit hours are available each day. The Upper and Lower regulating outlets have a 200 foot head restriction. This means that the Upper regulating outlets may be used when the pool is below elevation 1540 feet and the lower regulating outlets may be used when the pool is below 1465 feet.

When the reservoir is above spillway crest, elevation 1544 ft, (typically April 7 through September 21) changing spillway flow and generation may occur at any time; however, pre-scheduled transitions between outlets are preferred. Pre-scheduled outflow will determine how many unit hours are available each day.

After June 1 it is preferable to perform operational temperature control. Increasing spill to 100% will exceed downstream temperature targets which will disrupt adult Chinook migration and increase disease vectors between fish. It is preferred that spillway flow and generation be pre-scheduled with spillway flow occurring after dark (to cool off the water with the night air) and the mix of spill and generation not exceed 60% spill to 40% generation.

Fisheries considerations are as follows:

- When spill is required, maintain minimum spillway gate openings.
- Adjust spill to reduce total dissolved gas below 110%. Elevated TDG may impact survival of incubating/sac-fry spring Chinook salmon during the fall through spring.

- Operational Temperature Control is an annual operation until an actual temperature control tower is constructed. Operational temperature control begins on June 1 and ends when the reservoir drafts below spillway crest. The mix of spill to generation flow water is not to exceed 60% spill.

  In October when Chinook salmon eggs are incubating and we are exceeding the recommended temperature limits, the generation flow will be mixed with upper Regulating Outlet flow to reduce the downstream temperature. The mix of spill to generation flow water is not to exceed 50% spill. If the pool drafts below elevation 1465 feet the lower regulating outlets may be opened to release the coldest water stored in the reservoir. This may continue until the pool turns isothermal (one uniform temperature caused by a fully mixed body of water), typically in mid to late November.

- **Lookout Point**
  When the reservoir is below the operational limit of the spillway, elevation 890 ft, (typically October 15 through March 15) the project may change between RO flow and generation at any time; however, pre-scheduled transitions between outlets are preferred. Pre-scheduled outflow will determine how many unit hours are available each day.

  When the reservoir is above the operational limit of the spillway, elevation 890 ft, the project may change between spillway flow and generation during normal staff hours of 0500 – 1700 Monday through Friday or at shift changes at 0400 and 1600 Saturday and Sunday. Pre-scheduled outflow will determine how many unit hours are available each day.

  Functional use of the RO is only available when the reservoir is below elevation 900 ft due to a head limitation on the equipment.

  From June 1 until the weather warms up spillway flow and generation may be exchanged as needed on the hours specified above in the pre-scheduled release.

  Caveats to this operation include:
  - Spill through spillbay 4 is preferred. Spill through spillbay 3 is preferred after spillbay 4 is opened.
  - Spillway gates must be manually opened.
    - One way to avoid manual gate openings at Lookout Point would be to install remote control of two or more of the spillway gates and add camera(s) that will let the operator observe the forebay in front of these gates prior to remotely opening them.

  Fisheries considerations are as follows:

- When transferring flow from generation to 100% spill, spillbays 4 (preferred) and 3 or Regulating Outlets 4 (preferred) and 3 should be used to avoid elevated TDG. High TDG can impact the Dexter Fish Facility.

3. Projects with **very little to no flexibility** include Cougar, Big Cliff, Dexter, Foster and Lost Creek. The preferred operational choices for these projects are described below.

- **Cougar**
  No RO use per priorities due to gas cap limitations and operations of the Cougar Adult Fish Trap. Use the ROs for unit outage only.

  Fisheries considerations are as follows:
  - Turbine flow provides far-field attraction to the Cougar Adult Fish Trap (May through October).
  - RO flow over the gas cap causes TDG impacts on incubating/sac-fry spring Chinook salmon located below Cougar Dam.
  - Elevated TDG may impact spring Chinook fry present in spring.

- **Big Cliff**
  No voluntary spillway flow per priorities. Spillways should be used for unit outages only. This project does not have ROs and therefore cannot operate below the spillway crest except while generating.

  Fisheries considerations are as follows:
  - All life stages of spring Chinook salmon and winter steelhead are present below Big Cliff Dam.
  - Salmonid fry are present below the projects from fall through spring. This is the life stage most susceptible to ramp related stranding.
  - Elevated TDG may impact survival of incubating/sac-fry spring Chinook salmon during the fall through spring.
  - Steelhead fry are present through July below Big Cliff Dam and are highly susceptible to stranding from down ramping that exceeds targets.
  - Big Cliff Unit assists in degassing to reduce TDG levels.

- **Dexter**
  No voluntary spillway flow per priorities. Spillways should be used for unit outages only. This project does not have ROs and therefore cannot operate below the spillway crest except while generating.

  Fisheries considerations are as follows:
  - Salmonid fry are present below the projects from fall through spring. This is the life stage most susceptible to ramp related stranding.
  - Elevated TDG may impact survival of incubating/sac-fry spring Chinook salmon during the fall through spring.
  - Dexter Unit provides far-field attraction to the Dexter Adult Fish Trap.

No-Load, No Generation Flexibility for NWP non-Mainstem Columbia Hydropower Projects

- **Foster**
  No voluntary spillway flow per priorities. Spillways should be used for unit outages only. This project does not have ROs and therefore cannot operate below the spillway crest except while generating.

  Fisheries considerations are as follows:
  - All life stages of spring Chinook salmon and winter steelhead are present below Foster Dam.
  - Salmonid fry are present below the projects from fall through spring. This is the life stage most susceptible to ramp related stranding.
  - Elevated TDG may impact survival of incubating/sac-fry spring Chinook salmon during the fall through spring.
  - Steelhead fry are present through July below Foster Dam and are highly susceptible to stranding from down ramping that exceeds targets.

2.3 Special Operations Requests

A Special Operation is one that is outside of normal or typical operation for a project. It may or may not be outside the approved Water Control Plan. Special Operations could include such things as changes to specific release targets, specific pool levels, specific river stages, flow targets, water temperature targets, limitations on outlet use, or other non-routine operations that are important to the overall benefits of the authorized purposes for Corps projects.

Special operations are one-off or short-term. Permanent changes to operations may require additional documentation and approvals. NWP has developed a procedure for requesting and tracking approvals and implementation of Special Operations. An online site (Sharepoint) has been established to submit, review, and approve requests. Regulators should ensure that special operations have been approved by the Chief, Reservoir Regulation and Water Quality Section prior to scheduling the operation. Additional information on the procedure for requesting special operations can be found in the *Special Operations Request and Implementation Procedure* document.

Special Operations Requests are located on the Sharepoint site.

https://team.usace.army.mil/sites/NWP/EC/H/water_mgmt/default.aspx

3 Conservation Season Operational Notes

Conservation Season is from February through October, including filling season and release season. Reservoir levels may vary between mid-April and early September, depending on inflow to the reservoir, required releases for authorized purposes and surface evaporation. In order for the reservoir to fill for maximum summer conservation use, sufficient seasonal runoff must occur to balance or overcome the required releases and evaporation.

Once water reaches the highest achievable reservoir level during the filling period, the Corps operates the storage reservoirs for conservation purposes, including recreation and environmental uses. The projects are generally kept as full as possible; however, operations to meet tributary and mainstem flow targets specified in the Willamette Biological Opinion, combined with surface water evaporation during the drier summer months, typically result in a "draw down" over the course of the summer.

At Fern Ridge, there are legal requirements to release a minimum amount of water into the Long Tom River downstream of the reservoir. These releases help maintain water quality, support fish and wildlife and provide for purchased irrigation water. Lookout Point, Cougar and Detroit are also operated to meet release temperature targets. The current targets are those developed for the McKenzie by the Resource Agencies in the 1980s.

3.1 Conservation Plan Development

The augmentation of flows in the Willamette Basin is a coordinated effort among the state, local and federal agencies concerned with the flow conditions. The Corps takes the lead, through EC-HR, to put together a Willamette Basin Project Conservation Release Season Operating Plan (Conservation Plan), which is shared with interested agencies.

Operational planning for the conservation release season begins with the February water supply forecast and continues through October. The Conservation Plan identifies flow and storage needs for each tributary and Corps reservoir in the Basin based on the anticipated total system storage in mid-May, from the April forecast. For more information on the Conservation Plan development process, please refer to the National Marine Fisheries Service (NMFS) 2008 Biological Opinion (BiOp), Appendix B, Section 5. The BiOp listed minimum mainstem flows, tributary flows and ramp rates along with the priority of reservoir drawdown for mainstem flow augmentation is provided in Section 8.

The basic steps to develop the annual Conservation Plan are as follows:

1. Beginning in February, evaluate the Natural Resources Conservation Service (NRCS) monthly water supply forecast for each project. When the monthly forecast is not available, runoff information can be supplemented by the ESP (Ensemble, also known as Extended Stream flow Prediction System) forecasts from the National Weather Service. These forecasts help estimate how full the reservoirs may be in May and June and how much stored water will be available for release during the conservation season.
2. Currently EC-HR runs a HEC-ResSim model to evaluate the water supply forecasts. This is the "long-term model" that is used for planning. During February and March, the focus is on refill.

The focus transitions to planned releases for the conservation season when evaluating the April, May, and June forecasts.

3. Interagency meetings are held each month, beginning in February. The water supply forecast is discussed as well as observed hydro-meteorological conditions.
4. After the mid-May project storage is realized, the plan is close to being finalized. The June NRCS water supply forecast is analyzed and if it doesn't vary significantly from the May forecast the plan is finalized and sent to the agencies and public. Previous conservation plans are stored at: \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Willamette_Basin\Regulator_Information\Water_Year_Folders\AARs Each year the conservation plan changes to include the specific year's water year type, minimum mainstem and tributary flows, and upcoming special operations that affect storage. Refer to the 2008 BiOp, Appendix D, Section 5, Willamette Conservation Plan Development, for further information regarding differentiating management tactical approaches.
5. Although the plan is presented as "final," changes can be made throughout the conservation season to ensure that benefits to the authorized project purposes are maximized. Changes should be coordinated with the WATER Interagency Group.

3.2 Drought Year Operational Notes

A drought year, or deficit year per the NMFS 2008 BiOp, will be designated when the April water supply forecast projects the Willamette Valley system storage by mid-May at or below 0.90 MAF. In a deficit year, the annual Willamette Conservation Plan development process will likely need to determine minimum flow objectives that are below the biological and/or Congressionally authorized minimum flow objectives provided in Section 8.

Drought Contingency Plans for each project are included as an exhibit to the Water Control Manuals. At the time of this writing, the Drought Contingency Plan is in DQC Review. The file is located here: \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Water_Control_Manuals\DroughtContingencyPlan

For guidance on operational decisions made during a recent drought, refer to the 2015 Drought Year After-Action Report:
\\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Past_Events\WY2015Events\2015_Drought_AAR

3.3 Spring and Early Summer Operational Notes

May and June can still bring significant runoff through rain showers when the projects are full or close to full. Most projects have a limited amount of storage above maximum conservation pool so it is very important to keep an eye on the spring weather.

Opportunities to release environmental flows (e-flows) should be monitored. Implementation of e-flows is event-driven and left to the regulator's/operator's judgment. For the specific procedures of implementing e-flows, refer to the updated water control manuals for CGR, DET, and LOP.

\\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Water_Control_Manuals\Add_Eflow_Memo

Below are additional notes on operating the projects this time of year.

Hills Creek:

- Essentially pass inflow and maintain pool between 1540.0 and 1541.0 ft. Watch the elevation of the pool and try to keep it below 1542.0 ft.
- Keep in mind that large increases in releases move downstream and turn into Lookout Point inflows.
- When balancing spring flow augmentation, it is better to keep Hills Creek full and augment from Lookout Point. There is more recreation at Hills Creek then at Lookout Point.
- Minimum release is 300 cfs, but ODFW prefers a minimum release of 400 cfs.

Lookout Point:

- Lookout Point is used to augment mainstem flows at Albany and Salem. Outflow typically ranges between 2,500 and 3,000 cfs and refill is not guaranteed during the spring.

Dexter:

- If spill is necessary, spill across as many of the 7 bays as possible to reduce Total Dissolved Gas (TDG) levels. See Section 7.1 for more information on use of the spillways.

Fall Creek:

- Pass Inflow as long as possible and maintain pool between 829.0 and 830.0 ft. Fall Creek is a widely used recreation project so try to keep the project as full as possible.

Cottage Grove:

- Pass inflow and maintain the pool between 789.0 and 790.0 ft. If the pool rises above 791.0 ft, the reservoir will free-flow over the spillway.

Dorena:

- Pass inflow and maintain the pool between 831.0 and 832.0 ft.

Fern Ridge:

- Pass inflow and maintain the pool between 373.3 and 373.5 ft. Fern Ridge is a widely used recreation project so try to keep the project as full as possible. There is also a lot of wildlife and plant life in the reservoir, so don't go above 373.5 ft since small changes in lake level can flood many acres of prime bird nesting habitat.
- Support minimum flows at Monroe. MNRO minimum flow is 50 cfs in May and June and 30 cfs in July and August for downstream irrigators.
- The Long Tom River part of the Fern Ridge flood reduction project. It is a long, diked canal with much seepage, so it does not help Salem flows to augment from Fern Ridge.

Blue River:

- Balance augmenting mainstem flows with the water stored behind Blue River with water stored behind other projects such as Cougar and Lookout Point. Don't draft Blue River first then move to another reservoir.

- Monitor the interim draft limits to ensure that there is enough stored water for late summer and fall minimum releases.

Cougar:

- The BiOp lists 300 cfs as the minimum flow but Jeff Ziller (ODFW) always requested flows between 400 and 450 cfs to keep water in some of the side channels. This also works well with turbine flow and attraction to the adult fish ladder.
- Early reservoir drawdown is necessary to achieve the low maximum outflow required by the BiOp of 580 cfs in September.
- It is good to target being slightly above the low water boat ramp, Slide Creek and Echo Park Ramps at elevation 1635 ft, after Labor Day to maximize recreation opportunities in the reservoir.
- E-flow implementation note: outflow above 5,000 cfs will inundate the Cougar adult fish collection facility water intake system, which include electrical and airburst equipment.

Green Peter:

- Stored water in Green Peter must be used cautiously to ensure that enough water remains in the fall to meet the spawning requirements. The outflows required by the BiOp are much higher than inflow in the summer and fall which makes the reservoir draft quickly. It is best to be full in May then stay on minimum outflow all summer. Interim draft limits and Res Sim modeling will help determine how much volume to use for mainstem augmentation.
- If the reservoir elevation is drafting close to the interim draft limits, the fall spawning minimum flow may be delayed until mid-September or until the fish show up. Coordination with NMFS is required.
- The GPR units are able to run down to minimum power pool elevation 887 ft without cavitation.

Foster:

- Fill Foster prior to Memorial Day weekend for recreation. Currently the Project prefers to not surcharge the pool above 637.5 ft since resources aren't available to sweep the pool of woody debris.
- After the fish weir operation, pass inflow and maintain pool between 635.0 and 637.0 ft.
- During the spring and summer time frame, try to limit flow increases out of Foster to 300 cfs per hour for protection of fisherman and flow decreases to 300 cfs per hour for protection of juvenile fish.

Detroit:

- Pass inflow for as long as possible while the maintaining pool between elevation 1563.0 and 1564.0 ft. Detroit is a widely used recreation project so try to keep the project as full as possible.
- Maintain the BiOp minimum flows.
- Notify management if forecast indicate dropping below 1546 ft (critical elevation for marinas) before Labor Day weekend.

Big Cliff:

- Outflow above 10,000 from BCL may inundate the road access to Fishermen's Bend trailer park, so the operators provide phone notice

prior to exceeding this flow. Outflows above 10,000 cfs may be implemented as part of normal flood operations.

3.4 Fall Drawdown Operational Notes

Fall drawdown doesn't necessarily begin on September 1st as suggested by the flood control rule curves. Projects will typically start drawing down in early summer as stored water is released for conservation purposes. Many projects will be below their maximum conservation pool by the end of August. When conditions allow, pool levels are kept as high as possible, up to the flood control rule curve, for Labor Day recreation opportunities. It is often prudent to send out a press release regarding fall drawdown.

If for some reason the projects are not drawn down through the conservation season, drawdown rates to maintain flood damage reduction storage is located in the WCMs. Below are additional notes on operating the projects this time of year.

Power Projects:
- During the fall drawdown, the power projects should not spill unless units are unavailable or a special operation has been coordinated.

Hills Creek:
- Project is usually drafted at full powerhouse capacity (1,500 cfs) or 925 cfs above inflow, whichever is more appropriate.

Lookout Point:
- Project drafts through the summer while augmenting to maintain the mainstem Willamette River flow.

Dexter:
- Maintain 2,500 cfs during September and October.
- Boat races can extend into September and typically require a flat pool elevation. In order to hold pool flat at 693.0 – 694.0 ft, run Lookout Point turbines at a reduced rate of generation (approximately 2,500 cfs/unit) for 24 hours and maintain steady release out of Dexter.

Fall Creek:
- Maximum release from September through mid-November is 1,000 cfs if possible. The State prefers 800 cfs to aid in juvenile passage.

Cottage Grove:
- Cottage Grove project is not very large, so it is important to stay close to the rule curve.
- Downstream irrigators do not expect flows to be increased until after Labor Day. Pump damage can occur if releases are increased while irrigators are still in water.
- Typically schedule a one foot operating range once the reservoir drafts to minimum conservation pool.

Dorena:
- There is Reed Canary Grass in the top 7 feet of the reservoir where many small fish reside (mostly catfish, bass, and perch juveniles). If the pool drops too quickly, the fish cannot find their way out of the grass. Try to limit drawdown to 0.5 ft/day until elevation 825.0 ft is reached. That corresponds to a flow of 436 cfs above inflow.

- Dorena project is not very large, so it is important to stay close to the rule curve.
- Typically schedule a one foot operating range once the reservoir drafts to minimum conservation pool.

Fern Ridge:

- This is the last project to begin drafting. Try to avoid starting the drawdown until a Monday or Tuesday after October 1 since farmers are still moving heavy farm equipment across the roadways until then.
- Christie Johnson (OD-V) likes to put out a public release a few weeks before outflows are increased to draft the project. The public release warns the recreational users, the marinas, and the downstream irrigators that flows will be increased. The press release usually indicates several target releases and target dates. It is implied that we usually do intermediate ramps to get at target releases. *For example we may be at minimum flow of 50 cfs with a target of 1,000 cfs by 5 October. After 1 October, we would ramp up 200 to 300 cfs per day, during daylight hours.* It is important to notify Christie Johnson what the flow increase schedule will be so that she can inform the irrigators.
- Typically schedule a 0.5 ft operating range once the reservoir drafts to minimum conservation pool.

Blue River:

- Drawdown does not need to start if project is below the rule curve. Time outflow increases with the intersection of the rule curve.
- Typically schedule a one foot operating range once the reservoir drafts to minimum conservation pool.

Cougar:

- Maintain 580 cfs through September and increase to full powerhouse capacity in October. See fisheries notes in Section 6 for information on the operation.

Green Peter:

- If the reservoir elevation is drafting close to the interim draft limits, the fall spawning minimum flow may be delayed until mid-September or until the fish show up. Coordination with NMFS is required.
- Increases are usually made in September to avoid going above 1,500 cfs out of Foster in September through mid-October. A gradual increase in flows works well so releases can be evaluated as the fall progresses.
- At times the rule curve may be intersected 1 to 2 weeks before Foster Reservoir drafts (October 15th). It works well to have at least one week between increasing Green Peter outflow and ramping up Foster outflow.

Foster:

- Maximum release of 1,500 cfs is the target in September and October.
- The project may be kept high through mid-October if the hatchery requests it. In the past few years, drawdown has been delayed until October 15th at which time drawdown is accelerated (675 cfs more than inflow) in order to catch up to the rule curve by around October

24th. After the 24th the normal drawdown rate becomes 270 cfs more than inflow. The reason for a higher pool is that the hatchery intake is at elevation 635 ft so that it can obtain warmer water for the hatchery. The warm water is essential for maximum growth rate for the juvenile fish.

- After the Foster pool is drawn down, try to maintain a daily elevation between 614 and 616 ft. This will ensure that the S. Santiam hatchery intake valve is kept under water. The lower valve entrance is at elevation 610 ft.

Detroit:

- A gradual increase in releases allows the project to stay a little above the rule curve for a few weekends beyond Labor Day, while keeping it safe for downstream users.
- The State (ODFW) has requested flows stay between 1,500 – 2,500 cfs from September to 15 October.

4 Flood Damage Reduction Operational Notes

The primary flood season usually lasts from November into February. However, some projects also have summer flood control storage available. The basic operation is to maintain the reservoir at or below the seasonal flood control rule curve unless regulating an event. During a high water event, projects use the available storage to store water and control releases to try to maintain downstream control points below the regulation goals (bankfull). The Willamette basin is known to be flashy and conditions can drastically change very quickly.

For small floods, which are the most common and do not threaten to fill the reservoir, regulation is fairly simple. The plan is to keep the control points below regulation goals and store the water in the projects. Project outflows are reduced as downstream control points approach the regulation goals. Outflows will remain at minimum flow until the downstream control points recede below regulation goals. Once flows recede, the project will increase outflow, without exceeding regulation goals at downstream control points, until all stored flood water has been evacuated.

For large floods, the operation is similar, but the downstream control points will most likely exceed the regulation goals and project outflow may not remain at minimum during the event. When a project is relatively high in the pool and inflows are still high, a special regulation schedule is used. The regulation schedule is presented graphically as a set of curves in the WCM. These “special curves” or “special regulation curves” permit a gradual increase of the reservoir release instead of a sudden large spill, such as would occur if the reservoir were to fill before the end of the flood using the normal release schedule. This is described in more detail in Section 4.3.

When the projects fill above their scheduled rule curve, the goal is to bring the project back down to the rule curve within 10 days without exceeding bankfull at downstream control points. This is achieved by drafting at the normal maximum release rate while staying below regulation goals at the control points. This is not always possible as new storms can produce higher inflows and discharge more water downstream of the projects. If benign weather is forecast, it may be possible to draft the reservoir more slowly than if back-to-back events are expected.

4.1 Flood Operations

Flooding in winter is from rain-snowmelt events. The basic procedure is to pass inflow through the projects until it is no longer possible without exceeding the flood regulation goals downstream. At this point, releases will be adjusted to try to keep the control points below the regulation goals. The basic steps are as follows:

1. Collect data from the gages.
2. Compute lake inflow
    a. See Section 11.4 for calculating inflow based on change in reservoir storage.
3. Compare observed flows at the downstream control points to the regulation goals.
4. Compute target release required for downstream regulation goals. Regulation goals at the downstream control points are provided in Table 2 in Section 7.

a. The projects will typically release discharges equal to inflow until it is no longer possible to continue passing inflow without exceeding regulation goals at the downstream control points.
b. Reduce outflow as much as needed, down to the minimums specified in Table 12 in Section 8, before the flood regulation goals are predicted to be exceeded. Each project's Water Control Manual, along with other references, lists the normal and maximum rate of increase and decrease for outflow changes.
c. Do not forget about travel time between the projects and the downstream control points when deciding about when to begin reducing outflow. Keep in mind that the winter travel time from Lookout Point to Salem is nearly two days. If a big storm event is forecast three days out, reductions may be prudent at the southern project before the "rain hits the ground".
d. A schematic of the Willamette Basin projects, including downstream regulation goals and winter travel times, is shown in Figure 24 in Section 7.

5. Check the special curves to see if they apply. Based on observed inflow and pool elevation, outflow may need to be increased before the downstream control points begin to recede to make maximum use of the remaining storage in the pools. The special curves are described in Section 4.3.
6. Check project elevations to see if they are approaching the Interim Risk Reduction Measure (IRRM) elevations. If so, refer to the Water Control Plans for further regulation guidance.
7. Determine the required release for flood damage reduction based on Steps 4-6 above.
8. When downstream flows recede below the regulation goals, increase outflow to draft the reservoirs while keeping the downstream control points below the regulation goals.
9. When increasing and decreasing flow, follow the ramp rates and maximum and minimum flows shown in Table 12.

Opportunities to release environmental flows (e-flows) should be monitored. Implementation of e-flows is event-driven and left to the regulator's/operator's judgment.

For background information go to

\\nwd\nwp\ETDS\Willamette\Floodplain_Restoration\Sustainable_Rivers_Project

Reports are located at

\\nwd\nwp\ETDS\Willamette\Floodplain_Restoration\Sustainable_Rivers_Project\References\SRP Workshop Reports

E-flow guidance was also included in the water control manual updates for Cougar, Detroit, and Lookout Point. The updated water control manuals are located at

\\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Water_Control_Manuals

4.2 Project General Considerations during High Flows

Here are some general *considerations* to remember during high water:

Large Power Projects:

- Do not release outflow via the RO or spillway above powerhouse capacity if the project is in the secondary flood control space (typically the bottom 25% of the storage) at HCR, LOP, GPR, & DET.

Detroit:

- For releases above 10,000 cfs, notify ODFW so they can prep for impacts to the Minto Fish Facility. Also, Fishermen's Bend trailer park, located a few miles downstream of Big Cliff, may have problems when BCL releases are more than 10,000 cfs. When releases are above 10,000 cfs, no homes are damaged, but the access road may be flooded. Usually when the flows rise, residents from Fishermen's Bend call the operators for information on flows. The operators have been forwarding calls to the Portland District PAO office.

Big Cliff:

- Project does not have active flood control storage. There is a substantial local flow that can enter this project during high precipitation (up to 2,000 cfs).

Dexter:

- There used to be a dike downstream of Dexter that had problems when releases were above 15,000 cfs at Dexter. This dike has been repaired and there are currently no problems with it.

Dorena:

- Normal maximum release is 5,000 cfs, but if the pool is rising quickly and it looks like special curves might be necessary, increases can be made early to 7,000 cfs. Going up to 7,000 cfs early can help prevent going to 13,000 cfs (as required by special curves). A release of 5,000 cfs brings water to the banks of the downstream trailer park and 7,000 cfs puts water into the yards. No houses should be damaged with a 7,000 cfs release. If flows are going to be increased above 5,000 cfs, call Christie Johnson or Tamara Schroeder to have someone check on the status of the trailer park.
- If the project is not going to special curves, but you are having problems drafting, coordination can be done to take releases up to 6,500 cfs. The Lookout Point operators and management should be notified and included in discussions. This was done in November 1998 and did not cause any problems downstream; however care was taken to keep Goshen below bankfull levels.

Fern Ridge:

- Call Fern Ridge Environmental Stewardship Supervisor at the Project Office if we plan on exceeding 4,650 cfs at Monroe. Long Tom River capacity is 4,000 cfs downstream of Fern Ridge. Normal evacuation rate from Fern Ridge is 3,000 cfs, but outflows can be taken up to 4,000 cfs if needed after pre-coordinate with Dam Safety. Maximum regulating outlet (RO) release is 3,000 cfs.

- If the reservoir is above the elevation of the spillway crest, it is important to put at least 500 cfs through the RO and the rest through the spillway to minimize downstream erosion. RO releases up to 1000 cfs will allow the night/weekend operator remote operating capability since all SW gate changes are made manually on site.
- To minimize manual spillway gate changes, it is a good idea to split a flow of 3,000 cfs as follows: 1,000 cfs through the RO and 2,000 cfs through the spillbay resulting in 1,000 cfs in flexibility via remote RO control from LOP.
- Project staff should visually inspect the stilling basin, Clear Lake Bridge abutments, and downstream embankments when project flows exceed 2,500 cfs. It is recommended that inspections be done every other day during the high discharge period.
- Inflow and outflow gage readings may be suspect. The forebay gage changes slowly since the pool is so large and shallow, resulting in unreliable inflow calculations. The outflow readings are suspect at very high rates of discharge, although the USGS re-measured and updated the ratings in 2006 during high water. Vegetative encroachment along the channels may have complicated accurate gage readings, but as of 2008 most of the vegetation has been cleared.
- Critical constrictions along the Long Tom river include the bridge at Highway 36 at Cheshire, the Stroda structure and a park on the West bank at Monroe.
- *Example: Winter storms in January 2006 caused releases as high as 6,000 cfs and flow was split between the spillway and RO with a minimum RO flow of 500 cfs. The goal was to regulate Monroe below flood stage (9 ft or 6,800 cfs). While trying to evacuate stored flood water, Monroe was pushed to bankfull (8.5 ft or 5,700 cfs). FRNO outflow was more accurately reported via XConnect – the USGS reported lower flow which was incorrect. The NWS indicates that action stage at Monroe is 8.5 ft, flood stage is 9 ft, and major flood stage is 12 ft. 9.54 ft was reached on 1/18/06.*

Foster:

- Pass inflow from Foster and GPR for as long as possible while maintaining WTLO below bankfull (18,000 cfs).
- If it looks like WTLO will rise above bankfull, shut off Green Peter and pass inflow from Foster.
- If Foster inflow exceeds 10,000 cfs, begin storing into Foster and maintain releases at 10,000 cfs. If Foster fills too full and inflows are not receding, follow special curves.
- When inflows have dropped and downstream control points have peaked, draft Foster out at 12,000 – 15,000 cfs (whatever keeps WTLO at or below bankfull). If Green Peter is also quite full and there is another storm approaching, draft Green Peter and Foster concurrently. Remember that evacuation flows should not go above bankfull or result in a flow greater than that caused by runoff from the uncontrolled area during the previous flood. If the projects have used

less than 50% of flood control space, downstream points should not be taken to bankfull conditions while evacuating.
- When Foster has reached the rule curve, continue the 12,000 – 15,000 cfs and draft out Green Peter.
- Local flow to FOS can be estimated from the South Santiam at Cascadia (SSCO) and Wiley Ck (WNFO) gages whereas SSCO + ½ WNFO = FOS local inflow with a 2-hour travel time from the SSCO gage to the lake.

Green Peter:
- Inflow to GPR can be estimated from the Quartzville gage (QCCO) whereas 2xQCCO = GPR inflow with a 2-hour travel time from the gage to the lake.
- Balancing between GPR RO flow and GPR spillway flow during high water events can give the night operator more flexibility with flow changes. It allows the operator to change flows through the RO remotely since the spillway gate setting can only be changed during daylight hours when someone else is on site.

Hills Creek / Lookout Point:
- Hills Creek and Lookout Point should be kept near the same percent full during high water so that neither one is full while the other is empty.

Power Projects:
- When Detroit, Green Peter or Lookout Point is in secondary flood control space, we do not usually spill to get back to the curve. If the project is above the rule curve, but below the primary flood control point, draft at full powerhouse capacity.

Storage Projects with IRRMs:
- A special situation exists when a flood results in the pool filling to the Interim Risk Reduction Measure (IRRM) elevation or above. The projects have IRRMs in place during flood season that precludes operation of the spillway gates when the head on the gate is at the IRRM elevation (upper spillway gate strut in most cases) or higher. The pool will be allowed to fill above the IRRM elevation, but the spillway gates must be opened and tracking with the pool during filling. The special curves have been modified to incorporate this operation. See the Interim Water Control Plans for the current Flood Season for further regulation guidance. It should be noted that these IRRMs apply during flood season (November through February or as late into the spring as storms might indicate the need for spillway releases).

4.3 Special Regulation Schedule (Special Curves)

During a flood, or as a result of a series of closely spaced high flow events, the reservoirs can fill before inflows have dropped off. Without the special curves (special regulation schedules) the project might continue to release minimum flows to reduce downstream flooding but then suddenly need to increase releases to pass inflow as soon as the project reaches full pool. Instead, the special curves are used to determine the appropriate minimum releases from the project to keep the project from filling too high, too quickly, and to help protect downstream control points from abrupt high releases. The theory is to

gradually increase outflows to slow reservoir fill rate to avoid "fill then spill" situation. The curves are based on the elevation-storage relationship in the project and an analysis of typical large inflow hydrographs. Each project has its own set of special curves that can be found in the water control manuals. They are used as key guidelines to help determine appropriate releases. Special curves for each project can be found in their respective Water Control Manuals.

Each project has its own set of special curves that can be found in the water control manuals. They are used as key decision points to help determine appropriate releases. They can also be found at the following link: \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Willamette_Basin\Regulator_Information\Special_Curves

4.4 Flood Team Roles and Responsibilities

Several sections and agencies work together to achieve successful regulation and operation of the Willamette projects during a flood. These groups include EC-HR, project operators, Division RCC, National Weather Service (NWS), and NWRFC. Roles and responsibilities of some of those sections and agencies are described below:

- CENWP-EC-HR
    - Regulators make the release decisions using data, developing trends and the forecast for guidance.
    - Changes may be made every hour at the projects in response to changing conditions.
    - Will go to 24-hr operations if necessary.
    - In addition to the WM staff provides support through data collection and answering inquiries.
    - Provide weather, streamflow and reservoir briefings to the District.
- RCC
    - Also monitors flows and reservoir levels.
    - Available to provide input and help out with special situations.
    - Coordinates with EC-HR regarding deviations from rule curve.
    - Trained to provide backup regulation as needed.
    - Coordinates Columbia Basin System Flood situations.
- CENWP-EC-H
    - Includes Dam Safety, Hydrology and Hydraulic Design.
    - Available to provide input and help out with special situations.
- Project Staff and Project Operators
    - Support the regulators – they are the eyes and ears in the Willamette.
    - Provide EC-HR valuable feedback on actual field conditions.
    - Control rooms are staffed 24-hrs, so basin is watched at all times.
    - Often give the first warning call to EC-HR if inflows begin to rise.
    - Provide warnings to local emergency managers.
    - Project Office can provides field support during floods
- NWRFC

  - Forecasts river levels and the timing of peaks.
  - Updates flood forecasts as often as every 6 hours during a flood. Two to three times in a 24-hr period is more common.
  - NWRFC usually goes to 24-hour operations before EC-HR.
- NWS
  - Official entity that issues public forecasts and flood alerts (the Portland District Corps of Engineers **do not** issue flood notifications).
    - http://www.nwrfc.noaa.gov/
  - Issues flood warnings on controlled and uncontrolled streams, incorporating regulation decisions from EC-HR provided to the NWRFC.
  - The Portland Weather Service office issues official Willamette flood alerts.
  - For Willamette forecasts and other flood information the public is requested to call the Portland NWS office at (503)261-9246 or (503)261-9247.

5 Coordination

The close relationship between EC-HR and project operators is crucial to ensure effective and efficient operation and optimum flood reduction protection. In general, EC-HR has the responsibility for Willamette Basin system wide flood damage reduction decisions, conservation storage planning and scheduling, daily schedule development, and coordination of requests for special operations with other agencies. The operators are responsible for local flood damage reduction, adherence to project operating criteria, schedule implementation, project emergencies and forwarding requests to EC-HR for special operations.

EC-HR will pre-coordinate any operations related to Endangered Species Act, Total Maximum Daily Load (TMDL), fishery, hydropower or other special operations with the appropriate staff within NWP and agencies outside the Corps. EC-HR will also pre-coordinate flood control operations whenever possible. Communication is critical and EC-HR welcomes input from project operators regarding flood control operations, schedules, and schedule flexibility.

EC-HR will notify operators that schedules have been sent and operators will have the opportunity to respond with input. Operators should try to respond within one hour at which time schedules will be considered acceptable. Whenever possible, NWP should send schedules to the projects no later than 14:30 hours and any discussion or comments should be provided by 15:30 hours. EC-HR and project operators should use best efforts to coordinate or notify each other of changing conditions, which may affect previously coordinated schedules.

Heavy precipitation or rising inflows may necessitate a change in project release to provide optimum flood damage reduction. It may be impractical to send a schedule for each change. In this case, the project operators are to notify EC-HR of any changes so that flood control is taken into consideration. If time or conditions do not permit for schedule to be sent, EC-HR and project operators will decide on course of action considering local and system flood control.

Coordination of schedules can also be unofficial or official. For instance, EC-HR may call the projects to make adjustments to releases for expediency and then send out the scheduling on the CBT later as the official protocol calls for. During a storm, flow changes are often made over the phone and are ***not*** backed up with a teletype later due to frequent changes.

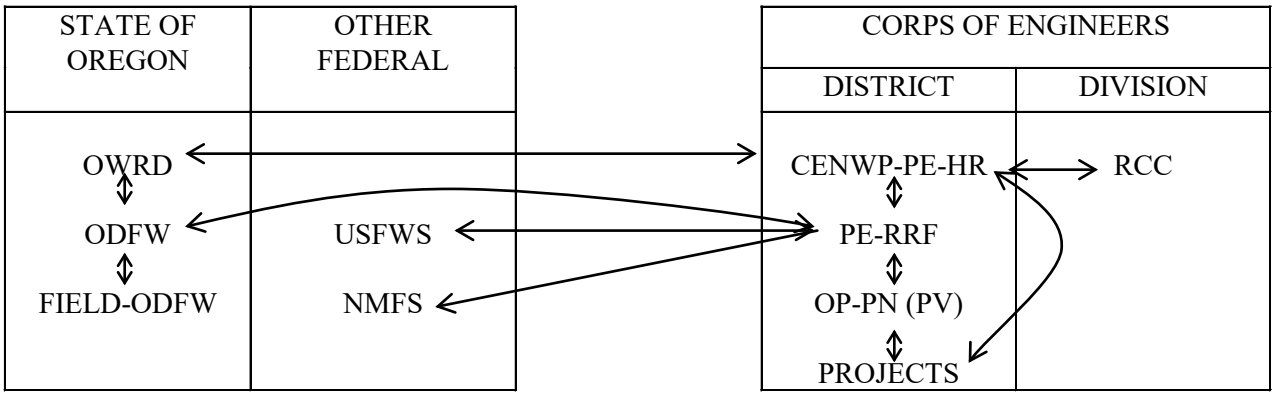
If an emergency is declared, operators are to follow project protocol and notify EC-HR of the situation. If during non-duty hours, consult duty list phone numbers of the regulator on duty. EC-HR will keep project informed of regulators on duty and numbers for cell phones.

5.1 Line of Command

There are official and unofficial lines of command. The official line of command is used for special operation requests from the public, outside agencies (except for outages from Bonneville Power Administration), or Division Reservoir Control Center (RCC) requests. For the Willamette Basin, coordination usually flows in the order shown in Figure 22.

Typically EC-HR contacts the State of Oregon including OWRD and ODEQ when necessary. OWRD then coordinates with ODFW. EC-HR also contacts Greg Taylor (Willamette Valley Project office at Lookout Point) or Ian Chane (OD-S) for fish regulations. Chris Wren (Lookout Point), Tom Voldbaek (Foster) and/or Dave Bardy (Lookout Point) should be contacted for coordination of special operations involving the Willamette projects.

Illustration of Line of Command.



5.2 Points of Contact

Points of contact (POC) for Willamette Basin operation and regulation periodically change. At this time of this writing, the list below contains the most up to date POCs. *Contact information for the positions listed below is available in the Outlook address book or the Portland District Reservoir Regulation Card that is updated annually.*

- Lookout Point Project Office
  Projects Operated – Hills Creek, Lookout Point, Dexter, Fall Creek, Cottage Grove, Dorena, Cougar, Blue River and Fern Ridge

Office: (541)937-2131
Control Room: (541)937-3072
Fax: (541)937-3401
Fax: (541)747-9408 (call before faxing)
Cougar Intake Tower: (541)822-1145

- Operating Project Manager – Erik Petersen
- Assistant Operating Project Manager – Dustin Bengston
- Assist. Project Engineer – Dave Bardy
- Chief of Operations and Maintenance – Greg Barrowcliff
- Environmental and Safety Officer – Rufus Faggons
- Fisheries Biologist – Greg Taylor
- Operators – DLL-CENWP LOP Operators (double click list name in outlook for current operations staff names)

- Detroit Project Office
  Projects Operated – Detroit and Big Cliff
  Control Room: (503)897-2385 or (503)854-3310 or (503)854-3150
  Fax: (503)854-3110
  - Chief of Operations and Maintenance – Tim Ernster
  - Operators – DLL-CENWP DET Operators (double click list name in outlook for current operations staff names)
- Foster Project Office
  Projects Operated – Green Peter and Foster
  Control Room: (541)367-5124
  Fax: (541)367-8446
  DATTS: 922-155
  - Chief of Operations and Maintenance – Tom Voldbaek
  - Operators – DLL-CENWP FOS Operators (double click list name in outlook for current operations staff names)
- Fern Ridge Dam
  Operated by Lookout Point
  Project Phone: (541)688-8147
  Fax: (541)688-8195
  - Supervisory Park Ranger
  - Environmental Stewardship Supervisor – Wendy (Gwendolyn) Jones
- Cottage Grove Dam
  Operated by Lookout Point
  Project Phone: (541)942-5631
  Fax: (541)942-1305
  - Park Rangers – Christie Johnson and Tammy Schroeder
- Reservoir Regulation (EC-HR)
  EC-HR Phone Line: (503)808-4896

*Phone numbers available in Outlook and the duty list and on the emergency contact card \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Water_Management\Emergency_Contact_Card

  - Chief – Salina Hart
    - Work: (503)808-4887
    - Work Cell: (503)819-4189
    - Cell: (503)720-0454
  - Primary Willamette Regulator – Mary Karen Scullion
    - Work: (503)808-4869
    - Cell: (503)927-4642
  - Backup Regulator – Yamen Hoque
    - Work: 503.808.4960
    - Cell: 503.819.9823
    - Personal cell: 503.601.9644
  - Backup Regulator – Kevin McAllister
    - Work: (503) 808-4967
    - Cell: (503) 318-3524
- Water Quality (EC-HR)
  - Primary –– Tina Lundell
  - Backup – Kathryn Tackley
- Data Collection and Transmission (EC-HR)
  - Primary – Arthur Armour
    - Work: (503)808-4833
    - Home: (503)228-2023
    - Cell: (503)757-3525
  - Backup – Ruth Abney
    - Work: (503)808-4958
    - Home: (503)901-9481
    - Cell: (503)913-1539
- Reservoir Control Center (PDW-R)
  - Team Lead – Aaron Marshall
    - Work: (503)808-3711
    - Home: (208)860-6137
    - Cell: (503)313-3319
  - Regulators – See Duty List
    -
- NWP Biologist
  - Primary – Greg Taylor (OD-V)
    - Home: (541)514-6497
- Marion County Public Works

  - Sam Steadman: (503)588-5304
- Kane's Marina
  - (503)854-3362
- Bill Sanderson Fishing Guide
  - (503)897-3301
- Santiam Water Control District
  - Larry Trosi: (503)769-2669
- Northwest River Forecast Center (NWRFC)
  Main Office: (503)326-7291
  - Hydrologist-In-Charge – Joe Intermill
    Joe.Intermill@noaa.gov
- National Weather Service – Portland Weather Forecast Office (WFO)
  Main Office: (503)261-9246
  24-Hour Phone Line: (503)326-2356
- Bonneville Power Administration (BPA)
  - Real-Time Desk – (503)230-4374 or (503)230-3341 (ask for Hydro Desk)
  - 3-Shift Desk – (503)230-5724
- United State Geological Survey (USGS) Gage Maintenance
  - James King (Lead)
    Office: 503-251-3462
    Work Cell: 971-275-3749
    jbking@usgs.gov
  - Jonathan Shannon
    Office: 503-251-3254
    Work Cell: 503-320-8194
    jshannon@usgs.gov
  - Jacob Flinn
    Office: 503-251-3460
    Work Cell: 503-730-9355
    jflinn@usgs.gov
- Oregon Water Resources Department (OWRD)
  - Western Region Manager – Bill Ferber (503)986-0903
- Oregon Department of Fish and Wildlife (ODFW)
  - Information: (503)872-5268
  - Headquarters – Dave Anderson: (503)872-5268
  - Jeff Ziller: (541)726-3515 ext.26
  - Long Tom – Steve Mamoyac: (541)757-4181 ext.249
  - Marian Forks Hatchery/Minto Fish Facility – Greg Grenbemer: (503)854-3522
  - South Santiam Hatchery – Bill Nyra: (541)367-3437
- Oregon Department of Environmental Quality (ODEQ)
  - Willamette Basin Coordinator-Nancy Gramlich (503)378-5073

gramlich.nancy@deq.state.or.us

- National Marine Fisheries Service (NMFS)
  - Stephanie Buchfield: (503)736-4720
- CEEIS 24-Hour Support Center
  Call when there is a problem with the Water Control Data System (CBT Messenger, Willamette Controller, etc.). Request they open a "WCDS Support Request" and indicate which system is failing and if the problem should be treated as urgent.
  - (503)808-5076 or (800)531-4472 and press "0"

- Useful Outlook Distribution Lists
  - DLL-CENWP FOS Operators
  - DLL-CENWP DET Operators
  - DLL-CENWP LOP Operators
  - DLL-CENWP Willamette Valley Operators
  - DLL-CENWP-EC-HR (NWP RCC)
  - DLL-CENWD-PDW-R (NWD RCC)

6 Fisheries Information

Waters within the Willamette Basin contain listed salmonoid species. The Willamette Project is being operated in accordance with two Biological Opinions issued by National Marine Fisheries Service and U.S. Fish and Wildlife Service. Some project specific fisheries information is listed below. Additional information can be found in the BiOps.

Dexter:

- There is a hatchery downstream of Dexter. In the spring (March) they may request 2000 cfs for a few days for hatchery smolt release.

Foster:

- The last remaining run of winter steelhead is above Foster Dam. For the last several decades the fill schedule has been modified to keep the Foster pool at elevation 614.0' (just above minimum conservation pool) from 15 April to 15 May. The low pool (drawdown) combined with spill has been relatively successful in moving juveniles downstream in the spring. Water is released from Green Peter over a few days to fill Foster before Memorial Day weekend.
- Currently we are involved in a study and the weir is operated continuously throughout the year. The summer operating range is 634-636 feet and the winter operating range is 615-617 feet.
- The South Santiam Fish Hatchery is located immediately downstream of Foster. They have special restrictions at certain times of the year for adults and juveniles regarding minimum and maximum flows.

Fern Ridge:

- Cutthroat trout live in the Long Tom River downstream of Fern Ridge.

Fall Creek:

- Chinook fingerlings are reared in the Fall Creek reservoir and are released each fall. In order for the fish to exit the reservoir safely, the draft schedule has been modified to draft the reservoir below Minimum Conservation Pool (to near 690.0 ft) by late fall (November). The low head combined with a maximum release of 400 cfs keeps the juveniles from hitting the far wall of the stilling basin and being injured. The water from the project can be jetted into the concrete wall at the end of the stilling basin and kill juveniles.

Blue River:

- Because Blue River is one of the first project to augment flows, the reservoir begins drafting early. A temperature control tower was authorized with the Cougar temperature control tower but never built.

Cougar:

- Listed Bull trout are in Cougar reservoir and downstream which means all special operations need to be coordinated. Spring Chinook migrate to the McKenzie so flows are restricted in the fall so the eggs won't be de-watered in the spring. An adult fish trap was constructed in 2010 and powerhouse releases are necessary to attract adult fish to the trap.

Detroit:

- The Minto fish collection facility downstream of Big Cliff was constructed in 2011-2012 and special operations for Minto will not be required. However, notify ODFW if flows will exceed 10,000 cfs.

7 Project Information

Basic project information is provided in the text, figures and tables below. Additional information is available in a number of places including the WCM, the Columbia River Tributaries Review Study Number 49 (CRT 49), and the pertinent data sheets stored on the LAN. Dam Safety Section (EC-HC) has also prepared fact sheets with project information that are available from the EC-HC website.

Table 2. Flood regulation goals at Willamette projects.

Station ID	Station Name	Action Stage/(Bankfull)[1,3]	Flood Stage[2,3]	Major Flood Stage[2,3]
GOSO	Coast Fork Willamette River near Goshen	11.7 ft (12,100 cfs)	13.0 ft (14,900 cfs)	18.0 ft (41,000 cfs)
JASO	Middle Fork Willamette at Jasper	9.4 ft (20,000 cfs)	10.0 ft (23,000 cfs)	15.0 ft (65,200 cfs)
EUGO	Willamette River at Eugene	20.2 ft (39,500 cfs)	23.0 ft (52,600 cfs)	29.0 ft (94,300 cfs)
VIDO[4]	McKenzie River near Vida	8.0 ft (22,200 cfs)	11.0 ft (35,000 cfs)	14.0 ft (49,500 cfs)
HARO[4]	Willamette River at Harrisburg	10.8 ft (39,700 cfs)	14.0 ft (66,500 cfs)	17.0 ft (100,700 cfs)
MNRO[4]	Long Tom River at Monroe	8.5 ft (5,660 cfs)	9.0 ft (6,780 cfs)	12.0 ft (16,000 cfs)
ALBO	Willamette River at Albany	21.6 ft (67,300 cfs)	25.0 ft (84,000 cfs)	32.0 ft (152,600 cfs)
WTLO	S. Santiam River at Waterloo	10.2 ft (19,000 cfs)	12.0 ft (25,700 cfs)	16.0 ft (42,700 cfs)
MEHO[4]	N. Santiam River at Mehama	8.9 ft (17,000 cfs)	11.0 ft (30,500 cfs)	13.5 ft (53,600 cfs)
JFFO	Santiam River at Jefferson	13.0 ft (43,000 cfs)	15.0 ft (55,900 cfs)	23.0 ft (213,000 cfs)
SLMO	Willamette River at Salem	21.2 ft (94,000 cfs)	28.0 ft (154,300 cfs)	32.0 ft (201,700 cfs)

[1]Action Stage [formerly "bankfull"] is set by the National Weather Service. It is defined as an established gage height at a given location along a river or stream, above which a rise in water surface will cause the river or stream to overflow the lowest natural stream bank somewhere in the corresponding reach. Refer to the new rating tables to determine flows since the ratings change on a regular basis, thus affecting flow.

[2]Flood Stage is set by the National Weather Service. It is defined as an established gage height for a given location above which a rise in water surface level begins to create a hazard to lives, property, or commerce. The issuance of flood advisories or warnings is linked to flood stage.

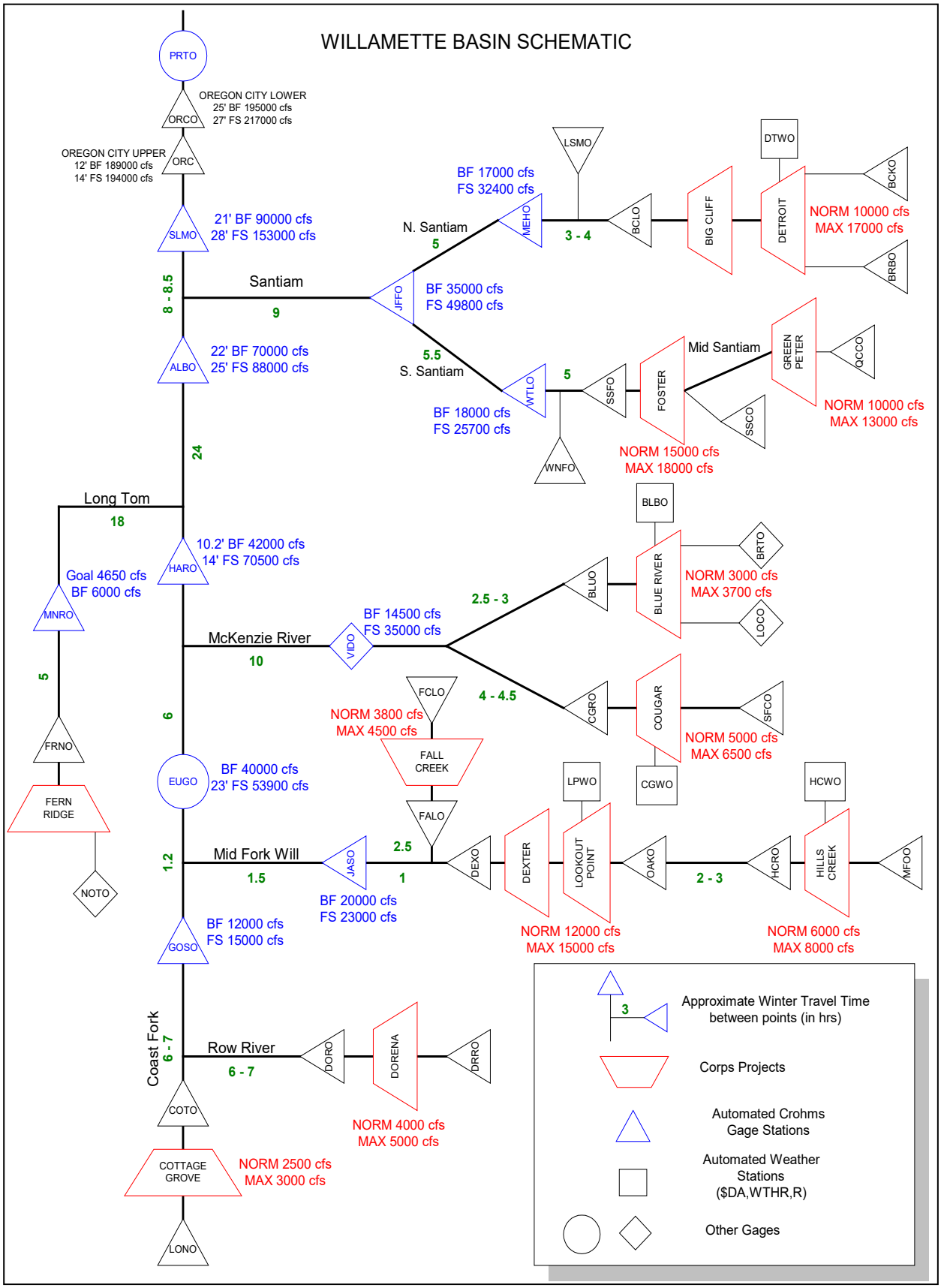
[3]Flows associated with Action Stage, Flood Stage, and Major Flood Stage may change as rating tables are updated.

[4]Action stage and regulation goal differ. VIDO regulation goal is 14,500 cfs. Harrisburg regulation goal is 52,000 cfs. Mehama regulation goal is 17,000 cfs. Maximum evacuation rate from Fern Ridge is 3,000 cfs.

Update – Oct 2018 – Table 2 but not Fig 24 as this image cannot be edited.
The regulation goal at Mehama should remain at 17,000 cfs and not the flow associated with the NWS Action Stage of 8.6 ft which is 15,600 cfs (rating curve 13 June 2018). The stage at 17,000 cfs is 8.9 feet.
The damage discharge curve from economics section shows damages start between 17,000 and 17,500 cfs.
However... this is based on 1985 level of development. That explains the 17,000, and I will say that in the WCM

The regulation flow goal at Salem should change according to the updated USGS rating curves and NWS Action Stage, therefore, it is 21.2 feet and 94,000 cfs (rating curve 28 Feb 2018).
SOP shows flow at NWS Action Stage at Salem is 91,000 cfs.
Damage curve shows damages (1972) start between 90,000 and 90,500 cfs with $1,200 of damages at 94,000 cfs.

The regulation goal at Jefferson should change according to the most up to date USGS rating and NWS Action Stage, 13.0 feet and 43,000 cfs (rating curve 23 May 2018) .
SOP shows flow at Action stage of 43,020 cfs (small difference).
Damage curve, based on 1972, shows damages start between 37,000 and 37,500, with $5,000 in damages at 43,000.



Schematic 1: Willamette Basin schematic including regulation goals at control points and travel times.

Table 3. Reservoir and outlet work stats (updated July 15, 2019 to show the elevation at the top of each spillway gate when closed).

Willamette Basin Projects
Reservoir and Outlet Works Stats

7/15/2019	HCR	LOP	DEX	FAL	COT	DOR	FRN	CGR	BLU	GPR	FOS	DET	BCL
Normal Evacuation Rate (cfs)	6,000	12,000	12,000	3,800	2,500	4,000	3,000	5,000	3,000	10,000	15,000	10,000	10,000
Maximum Evacuation Rate (cfs)	8,000	15,000	15,000	4,500	3,000	5,000	4,000	6,500	3,700	13,000	18,000	17,000	17,000
Elevation Levels (ft)													
Top of Dam (ft)	1548.0	941.0	702.5	839.0	808.0	865.0	382.0	1705.0	1362.0	1020.0	646.0	1579.0	1212.0
Max Pool (ft)	1543.0	934.0	697.4	834.0	802.6	860.0	375.1	1699.0	1357.0	1015.0	641.0	1574.0	1210.0
Full Pool (ft)	1543.0	929.0	695.0	834.0	791.0	835.0	375.1	1699.0	1357.0	1015.0	641.0	1569.0	1206.0
Max. Conservation Pool (ft)	1541.0	926.0	695.0	830.0	790.0	832.0	373.5	1690.0	1350.0	1010.0	637.0	1563.5	1206.0
Top of spillway gate when closed (ft)	1545.0	929.0	697.0	834.0			375.5	1699.0	1357.0	1015.0	641.0	1572.0	1206.0
Spillway Crest (ft)	1495.5	887.5	660.0	791.6	791.0	835.0	358.5	1656.8	1321.0	968.7	596.8	1541.0	1161.5
Min. Conservation Pool (ft)	1448.0	825.0	690.0	728.0	750.0	770.5	353.0	1532.0	1180.0	922.0	613.0	1450.0	1182.0
Min. Power Pool	1414.0	819.0	690.0					1516.0		887.00	609.0	1425.0	1182.0
Storage Levels (ac-ft)													
Top of Dam (kaf)	367.5	>504	-	132.5	-	-		-		446.9	-	-	7.320
Max Pool (kaf)	355.5	472.9	29.7	123.2	47.5	131.1	111.4	200.0	89.5	428.1	60.8	467.0	7.020
Full Pool (kaf)	355.5	451.0	27.3	123.2	32.9	77.6	111.4	200.0	89.5	428.1	60.8	448.5	6.430
Max. Conservation Pool (kaf)	350.0	438.2	27.3	116.0	31.8	72.1	97.3	189.0	82.8	409.8	55.9	428.8	6.430
Spillway Crest (kaf)	242.2	287.9	3.3	59.0	32.9	77.6	12.3	151.20	58.0	275.8	17.9	355.3	1.770
Min. Conservation Pool (kaf)	155.4	113.6	22.5	9.5	3.1	7.1	2.8	52.2	4.0	159.9	31.1	148.3	3.500
Min. Power Pool	106.7	102.2	22.5					43.5		97.3	27.50	111.9	3.500
Power Pool Storage (kaf)	48.7	11.4	0.0					8.7		62.6	3.6	36.4	0.000
Is Spillway Used?	No	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Number of Spillbays	3	5	7	2	1	1	6	2	2	2	4	6	3
Capacity of 1 bay at Full Pool (cfs)	42,500	41,862	35,400		40,800	0	(2@ a time)			-	41,100	24,290	59,670
Total Capacity at Min Cons. Pool (cfs)	-			-	-	-	-	-	-	-	40,000	-	-
Total Capacity at Max Cons. Pool (cfs)	130,000	194,000	-	70,000	-	-	45,000		37,500	92,500	170,000	98,580	-
Total Capacity at Full Pool (cfs)	141,500	222,000	242,000	82,500	40,800	0	45,000		53,200	110,000	200,000	145,740	179,000
Total Capacity at Max Pool (cfs)	141,500	270,000	267,000	82,500		97,500			53,200	110,000	200,000	191,640	179,000
Spillway Crest Elevation	1495.5	887.5	660.0	791.6	791.0	835.0	358.5	1656.8	1321.0	968.7	596.8	1541.0	1161.5
Date SW Crest reached on RC	26-Feb	12-Mar		3-Mar			8-Feb	7-Apr	3-Apr	27-Feb		4-Apr	
Number of Regulating Outlets	2	4	0	2	3	5	4	2	2	2	0	4	0
Size of RO	6.5*12.5	6'9"x12'		5'6"x10'	3'9"x6'6"	5'x6'	6'9"x9'8"	6'6"x12'6"	4'9"x8'	5'5"x10'		5'8"x10'	
1 RO @ 10% Open @ Min Cons Pool	300	525		250	83	105	130	340	150	426		520	
1 RO @ 10% Open @ Max Cons Pool	570	745		400	126	176	190	720	327	525		660	
1 RO @ 80% Open @ Min Cons Pool	2,415	3,240		1,930	611	754	955	2,640	1,243	3,600		3,165	
1 RO @ 80% Open @ Max Cons Pool	4,400	4,660		3,120	970	1,336	1,715	5,335	2,773	4,460		5,600	
1 RO @ Full Open @ Min Cons Pool	2,880	4,075		2,440	810	1,077	1,140	2,900	1,855	5,280		4,110	
1 RO @ Full Open @ Max Cons Pool	5,380	6,030		3,860	1,277	1,828	2,065	5,900	4,135	6,540		7,300	
Invert elevation	1,408.75	724		670	719	739	340	1,479	1,132	745		2@1,340&1265	
Max Useable Elevation		900										1,465	
Number of Turbines	2	3	1	0	0	0	0	2	0	2	2	2	1
Nameplate capacity at full pool (MW/unit)	15	40	15					14		40	10	50	18
Capacity per Turbine at Min Pool (cfs)	850	2,700	4200	-	-	-	-	690	-	2210	1710	2480	3200
Capacity per Turbine at Max Pool (cfs)	750	2,200	4,200	-	-	-	-	455	-	1,800	1,330	1,950	2,810
Total Cap. at Full Load at Min Pool (cfs)	1,700	8,100	4,200	-	-	-	-	1,380	-	4,420	3,420	4,960	-
Total Cap. at Full Load at Max Pool (cfs)	1,500	6,600	4,200	-	-	-	-	910	-	3,600	2,660	3,900	-
Intake, bottom, lake side, Invert elev (ft)	1,384	780	634.0	-	-	-	-	1,253	-	795	583.3	1,395.5	1,130
Top of Secondary Flood Control (ft)	1479.2	856						1568.9	1250	952		1484.5	
Top of Secondary Flood Control (kaf)	209.4	194.1						81.88	19.26	229.7		220.2	
Total Amount of Conservation Storage (kaf)	194.6	324.6		106.5	28.7	65.0	94.5	136.8	78.8	249.9	24.8	280.5	
	28%							22%				26%	
IRRM Elevation	1529							1685				1568.5	

Table 4: Boat ramp elevations by project. These elevations were re-confirmed by State Parks in 2011.

Hills Creek	
Boat Ramp	**Elevation**
Bingham Landing	1520 ft
CT Beach Park	1507 ft
Packard Creek	1441 ft
Lookout Point	
Boat Ramp	**Elevation**
Black Canyon	900 ft
Meridian Park	911 ft
Hampton Landing	911 ft
Signal Point	821 ft
Dexter	
Boat Ramp	**Elevation**
Lowell Park	684 ft
Dexter Park	684 ft
Fall Creek	
Boat Ramp	**Elevation**
Cascara Campground	822 ft
Winberry Creek Park	803 ft
North Shore Ramp	689 ft
Fern Ridge	
Boat Ramp	**Elevation**
Perkins Peninsula	368 ft
French Ridge Shores	367 ft
Richardson Park	365 ft
Orchard Point Park	364 ft
Cougar	
Boat Ramp	**Elevation**
Slide Creek	1635 ft
Echo Park	1635 ft
Blue River	
Boat Ramp	**Elevation**
Lookout Boat Launch	1330 ft
Saddle Dam	1295 ft
Dorena	
Boat Ramp	**Elevation**
Harms Park	820 ft
Baker Bay	765 ft
Cottage Grove	
Boat Ramp	**Elevation**
Wilson Creek	779 ft
Lakeside Park	745 ft
Green Peter	

Boat Ramp	Elevation
Whitcomb Creek	970 ft
Thistle Creek	919 ft
Foster	
Boat Ramp	**Elevation**
Calkins Park	631 ft
Gedney	619 ft
Sunnyside	613 ft
Detroit	
Boat Ramp	**Elevation**
Detroit Lake State Park Boat Ramp D	1556 ft
Mongold East Boat Ramp	1540 ft
Kane's Marina	1546 ft
Hoover Boat Ramp	1543 ft
South Shore Boat Ramp	1542 ft
Mongold Main Boat Ramp	1534 ft
Cove Creek Boat Ramp	1541 ft
Detroit Lake State Park Boat Ramp G	1530 ft
Mongold Low-Water Boat Ramp	1450 ft

7.1

7.2 Project Outlet Considerations

Many things have been learned about the projects through operations that are not necessarily identified in the design documents or the WCM. Below are notes on the operation of the outlets that have been passed on by operators to regulation staff.

Big Cliff:

Spillway: Spillway gates are operable from the Detroit control room.

Spillway: Spillbay 1 or 3 are set to auto open if the Big Cliff unit drops off line. If the operators need to spill to pass water, they would use the other two gates. Currently as of April 2017 spillbay 3 is set to auto open.

Turbines: One 18 MW unit available with a total outflow capacity of approximately 3200 cfs. This unit is remotely controlled from Detroit via SCADA.

Blue River:

Spillway: Spillway has never been used to pass water. There are erosion concerns in the spillway channel.

Regulating Outlet (RO): Regulating outlets are remote controlled from the LOP control room via SCADA.

Cottage Grove:

Spillway: Spillway is uncontrolled (free overflow over open weir).

Regulating Outlet (RO): Regulating outlets are remote controlled from the LOP control room via SCADA.

Cougar:

Spillway: Spillway has never been used to pass water. If used, the powerhouse access road would be washed out by flow from the spillway.

Spillway: Spillway gates are not remote controlled from the LOP control room. They must be manually opened and closed.

Turbines: Two 14 MW units available and controlled from LOP. They can be started and stopped remotely from LOP without on-site assistance.

Regulating Outlet (RO): Regulating outlets are remote controlled from the LOP control room via SCADA.

IRRM Limitation: 1685 ft during the winter months.

Detroit:

Spillway: Spillway gates can be remotely controlled from the DET control room.

Turbines: Regarding system emergencies, Detroit combined with Green Peter, with all units condensed could respond to a system frequency drop immediately with up to 180+ MW available depending on pool elevation. Two 50 MW units available co-located with control room.

Regulating Outlet (RO): Regulating outlets are remote controlled from the DET control room via SCADA. They shouldn't be used if the reservoir is above SW crest (1541 ft) unless they are approved for use (such as for temperature control in the summer if the units are out of service). The lower RO's can be if the reservoir is at elev 1465 or lower.

IRRM Limitation: 1568.5 ft during the winter months.

Dexter:

Spillway: Spillway gates (with the exception of gate 1) are operable from the LOP control room.

Spillway: Spillbay 1 is not remote controlled since it is the auto open gate if the Dexter unit drops off line.

Spillway: Spill across as many of the 7 bays as possible to reduce Total Dissolved Gas (TDG) levels while still maintaining the minimum gate opening.

Spill Through Individual Bays to Achieve Total Spill At Dexter (in cfs)							
Spillbay	**2,000**	**3,000**	**4,000**	**5,000**	**6,000**	**7,000**	**8,000**
1	-	-	-	-	-		
2	-	-					2,000
3	2,000	3,000	2,000	2,500	2,000	2,000	2,000
4	-		2,000	2,500	2,000	3,000	2,000
5	-				2,000	2,000	2,000
6	-	-	-				
7	-	-	-	-			

Turbines: One 15 MW unit is available and controlled from LOP.

Dorena:

Spillway: Spillway is uncontrolled (free overflow over open weir).

Regulating Outlet (RO): Regulating outlets are remote controlled from the LOP control room.

Turbines: A private company installed and are operating two turbines. For more information see the Dorena water control manual (Rev 2016). \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Water_Control_Manuals\Dorena

Fall Creek:

Spillway: Both spillbays released water in the spring of 2019, individually, to pass water from a spring storm that overtopped the spillway gates by nearly a foot of water. The spill gates, spill chute, and downstream spill channel all performed very well during the release. Downstream of the spillway is excellent habitat for Oregon Chubb and Western Pond Turtles due to beaver dams and small pools formed. The beaver dams were washed away in spring 2019 but we expect that they will be quickly rebuilt due to local permitting authorities for beavers

Regulating Outlet (RO): Regulating outlets are remote controlled from the LOP control room. In the spring 2019 operation the ROs were used in conjunction with the SW release enable smooth flow transitions downstream.

Fern Ridge:

Spillway: Spillway gates are not remote controlled from the LOP control room. They must be manually opened and closed. 2 spillbays are opened jointly (they are attached together by the gate control mechanics). So although there are 6 independent gates, you have 3 openings.

Regulating Outlet (RO): Regulating outlets are remote controlled from the control room. When water is above the spillway crest it is preferred to spill up to 2,000 cfs with RO outflow up to 1,000 cfs when total outflow is 3,000 cfs. This reduces erosion in the RO and gives the operator maximum flexibility to reduce RO outflow remotely. SW flow changes need to be made on-site. Maximum outflow is 3,000 cfs unless it is critical to draft at a higher rate – check special curves for guidance.

Foster:

Spillway: Spillway gates are operable from the Foster control room.

Spillway: Spillbay 1 is not operated since it is the auto open gate if the Foster unit drops off line.

Spillway: Spillbay 4 is used for the fish weir operation and is closest to the powerhouse and adult fish ladder.

Turbines: Two 10 MW units available co-located with control room.

Green Peter:

Spillway: Spillway gates are not remote controlled from the FOS control room.

Turbines: Regarding system emergencies, Detroit combined with Green Peter, with all units condensed could respond to a system frequency drop immediately with up to 180+ MW available depending on pool elevation.

Turbines: Green Peter is the only project that cannot ramp down from 2 units to 1 unit without condensing both units offline and then bringing them back to generate. This is because the turbines are above the tailwater and the water has to be drained out.

Regulating Outlet (RO): Regulating outlets are remote controlled from the control room. They may be used over the entire pool range!

Hills Creek:

Spillway: Spillway has never been used to pass water during a flood, but it has been tested, once. If used, the minimum spill through the bay closest to the powerhouse is 4,500 cfs. This flow is required to provide sufficient energy to push the flow past the powerhouse and keep the flow in the spillway channel. If the flow is less the powerhouse is flooded. Any required flow ramping will occur using the regulating outlet prior to transitioning from the regulating outlet to the spillway. EC-HD has more information on the spill test performed in the 1980s.

Spillway: Spillway gates are not remote controlled from the LOP control room.

Turbines: Two 15 MW units available remotely operated from LOP. Assume that they can NOT be started and stopped remotely. Operators must be on site to water up a unit that was shut down prior to starting it up. Operators can turn off a unit remotely from LOP via SCADA. If the unit is at SNL the unit may be started remotely from LOP. (The water-up valves are not remote controlled like they are at Cougar).

Regulating Outlet (RO): Regulating outlets are remote controlled from the LOP control room.

IRRM Limitation: 1529 ft during the winter months.

Lookout Point:

Spillway: Spillway gates are not remote controlled from the LOP control room. Manual gate changes are made at shift change (0400 & 1600 hrs) or during normal day shift.

Turbines: Two 40 MW units available co-located with control room.

Regulating Outlet (RO): Regulating outlets are remote controlled from the control room. Due to limitations on the walker valves, the RO's are used only if the reservoir is below elevation 900 ft.

7.3 Water Control Diagram (and Flood Control Rule Curve)

Based on project authorizations, the Corps developed a water control diagram for the operation of each project. The water control diagram is a water year based depiction of the authorized or allocated space in the reservoir. The diagram reflects the anticipated uses of water in the basin and the Corps' legal responsibilities and limitations regarding operation of the reservoirs. This diagram and the associated operational goals, limitations and requirements are outlined in the WCM. It includes specific elevation levels such as Maximum Pool, Full Pool, and the Flood Control Rule Curve. Storage components are identified, such as power pools, conservation pool, flood storage, exclusive summer flood storage, etc.

For multipurpose projects, the water control diagram includes a flood damage reduction rule curve. Using hydrologic inflow analysis, the Corps designed the rule curve to manage for the 100-year storm event, which has a one percent probability of occurring in any year. The rule curve is an extremely important component for operational decisions. It shows the maximum elevation to which the Corps should operate a reservoir during the year, with the exception of real-time flood operations. The Corps may operate under the levels indicated by the rule curve.

The rule curve specifies maximum reservoir levels (unless regulating an event) throughout the year along with flood damage reduction space required, filling rates to reach maximum conservation pool , and fall draft rates to meet winter flood control pool. It also specifies primary and secondary flood control elevations where applicable.

Each Willamette project rule curve indicates that the water level in the reservoir must be maintained at minimum levels whenever possible from December through January for the purpose of flood damage reduction. This allows adequate space to store and manage water during wet winter months. Refill of the reservoirs can begin as early as February 1st with the goal of reaching maximum conservation pool by mid-April to mid-May. Refill can be delayed when high runoff is expected which provides additional storage for flood damage reduction operations. Each reservoir may fill at a rate no faster than shown in the rule curve, unless managing for downstream floods. In summer, the Corps begins gradually drafting the reservoir to meet the authorized conservation season purposes and regain capacity for flood damage reduction.

The Willamette rule curves, including observed inflow and precipitation levels, are at

Internal, which includes forecast information

http://nwp-wmlocal2.nwp.usace.army.mil/nwdp/nwp_teacup/www/willamette/

External, which does not include forecast information

http://www.nwp.usace.army.mil/Missions/Water.aspx

8 Flow and Temperature Requirements

The Willamette Project is being operated in accordance with two Biological Opinions issued by each of National Marine Fisheries Service and U.S. Fish and Wildlife Service. The BiOp listed minimum mainstem flows, tributary flows and ramp rates are shown in Tables 4-7 and 12 below. Often the minimum flows listed in the Water Control Manual and the BiOps will differ. The priority of reservoir drawdown for mainstem flow augmentation is listed in Table 8. This information is included and discussed in the Annual Conservation Plan.

Cougar project was retrofitted with a selective withdrawal tower and is operated to meet temperature targets specified by the Resource Agencies. Detroit project is operated to provide interim temperature control by blending flow over the spillway with flow through the turbines and regulating outlets in the Big Cliff pool. Temperature targets are shown in Tables 9-11.

Table 5: Minimum mainstem flow targets for all years with the exception of drought years.

	Albany	Salem	
Period	**Minimum Instantaneous Flow (cfs)**	**Minimum Weekly Flow Threshold (cfs)**	**Minimum Instantaneous Flow (cfs)**
April		17,800	14,300
May		15,000	12,000
1 – 15 June	4,500	13,000	10,500
16 – 30 June	4,500	8,700	7,000
July	4,500	6,000	
1 – 15 August	5,000	6,000	
16 – 31 August	5,000	6,500	
September	5,000	7,000	
October	5,000	7,000	

Table 6: Transition periods for reducing minimum mainstem flow thresholds.

Period	Transition flow at Salem
27 April – 3 May	17,800 – 15,000 cfs
28 May – 3 June	15,000 – 13,000 cfs
12 – 18 June	13,000 – 8,700 cfs
27 June – 3 July	8,700 – 6,000 cfs

Table 7: Ramp rates specified in the BiOps.

Period	Ramp Rate
Daytime Max Hourly Ramp Rate	0.2 ft/hr
Nigh Time Max Hourly Ramp Rate	0.1 ft/hr
Daily Max Ramp Rate	The lesser of 50% of flow or 0.5 ft at the downstream gage.

Table 8: Minimum and maximum flow objectives below Willamette Basin projects.

Willamette Minimum Authorized Flows

2008 NMFS Biological Opinion

Location	Date	Minimum Flow (cfs)	Remarks
Detroit/Big Cliff	1 Feb - 15 Mar	1000	rearing/adult migration
	16 Mar - 31 May	1500	steelhead spawning
	1 Jun - 15 Jul	1200	steelhead incubation
	16 Jul - 31 Aug	1000	rearing
	1 Sep - 15 Oct	1500	chinook spawning
	16 Oct - 31 Jan	1200	chinook incubation
Blue River	1 Feb - 31 Aug	50	rearing
	1 Sep - 15 Oct	50	Chinook spawing
	16 Oct - 31 Jan	50	chinook incubation
Cottage Grove	1 Feb - 30 Jun	75	
	1 July - 31 Oct	50	
	1 Nov - 31 Jan	inflow	
Cougar	1 Feb - 31 May	300	rearing
	1 Jun - 30 Jun	400	rearing/adult migration
	1 Jul - 31 Aug	300	rearing
	1 Sep - 15 Oct	300	Chinook spawning
	16 Oct - 31 Jan	300	chinook incubation
Dorena	1 Feb - 30 Jun	190	
	1 July - 31 Oct	100	
	1 Nov - 31 Jan	inflow	
Fall Creek	1 Feb - 31 Mar	50	rearing
	1 Apr - 31 Aug	80	rearing, adult migration (Jun)
	1 Sep - 15 Oct	200	chinook spawning
	16 Oct - 31 Jan	50	chinook incubation
Fern Ridge	1 Feb - 30 Jun	50	
	1 July - 31 Oct	30	
	1 Nov - 31 Jan	inflow	
Green Peter/Foster	1 Feb - 15 Mar	800	Rearing
	16 Mar - 15 May	1500	steelhead spawning
	16 May - 30 Jun	1100	steelhead incubation
	1 Jul - 31 Aug	800	Rearing
	1 Sep - Oct 15	1500	chinook spawning
	16 Oct - 31 Jan	1100	chinook incubation
Hills Creek	1 Feb-31 Aug	400	Rearing
	1 Sep - 31 Jan	400	Migration and rearing
Lookout Point/Dexter	1 Feb - 31 Aug	1200	Rearing
	1 Sep 15 Oct	1200	chinook spawning
	16 Oct - 31 Jan	1200	chinook incubation

Table 8:

Project hierarchy when augmenting for mainstem flow goals.

No Augmentation	**Primary Spring Augmentation (April – 30 June)**	**Summer/Fall Augmentation (1 July – 31 October)**
Fern Ridge	Green Peter (Outflow up to 3,000 cfs. Prioritize filling Green Peter to Full to ensure enough storage to fill Foster and to maintain the summer/fall Biop minimum flows)	Lookout Point (Dexter outflow up to 2700 cfs.)
Detroit (Above elevation 1546 ft is adequate for recreation through 7 September.)	Cougar (Above elevation 1635 ft. Up to 600 cfs flow through the RO unless above rule curve.)	Cougar (Above elevation 1635 ft by 7 September and elevation 1532 ft by 31 October.)
Foster	Lookout Point (Dexter outflow up to 2,700 cfs.)	Hills Creek
Dexter	Hills Creek (Dexter outflow up to 2,700 cfs)	Green Peter (Foster outflow of 1,500 cfs from 1 – 30 September and 1,100 cfs 1 October – 31 January.)
Big Cliff	Blue River (Above elevation 1330 ft.)	Blue River
	Fall Creek (Above elevation 780 ft.)	Fall Creek (Above elevation 780 ft.)
	Dorena (Above elevation 825 ft.)	Dorena (Above elevation 825 ft.)
	Cottage Grove (Above elevation 779 ft.)	Cottage Grove (Above elevation 779 ft.)

Table 9. Temperature targets downstream of Cougar Dam.

	Maximum		Minimum	
Month	**Degree F**	**Degree C**	**Degree F**	**Degree C**
January	40.1	4.5	40.1	4.5
February	42.1	5.6	41.0	5.0
March	42.1	5.6	41.0	5.0
April	45.1	7.3	43.2	6.2
May	49.1	9.5	46.0	7.8
June	56.1	13.4	51.1	10.6
July	61.2	16.2	54.1	12.3
August	60.3	15.7	54.1	12.3
September	56.1	13.4	52.3	11.2
October	49.1	9.5	47.1	8.4
November	44.1	6.7	43.2	6.2
December	41.0	5.0	41.0	5.0

Table 10. Temperature Targets downstream of Dexter and Foster Dams

	Minimum		Maximum	
Month	**Degree F**	**Degree C**	**Degree F**	**Degree C**
January	40.1	4.5	40.1	4.5
February	41.0	5.0	42.1	5.6
March	41.0	5.0	42.1	5.6
April	43.2	6.2	45.1	7.3
May	46.0	7.8	49.1	9.5
June	51.1	10.6	56.1	13.4
July	54.1	12.3	61.2	16.2
August	54.1	12.3	60.3	15.7
September	52.3	11.3	56.1	13.4
October	<50	10.0	<50	10.0
November	<50	10.0	<50	10.0
December	41.0	5.0	41.0	5.0

Table 11. Temperature Targets downstream of Big Cliff Dam effective June 1, 2017

	Minimum		Maximum	
Month	**Degree F**	**Degree C**	**Degree F**	**Degree C**
January	38	3.3	42	5.6
February	38	3.3	42	5.6
March	42	5.6	44	6.7
April	42	5.6	46	7.8
May	46	7.8	50	10.0
June	48	8.9	54	12.2
July	52	11.1	55	12.8
August	52	11.1	55	12.8
September	48	8.9	54	12.2
October	46	7.8	52	11.1
November	42	5.6	46	7.8
December	41	5.0	46	7.8

Table 12. General flow guidelines as specified in the water control manual and Willamette Biological Opinion.

General Flow Guidelines															
	High Flow Periods						**All other periods**							Minimum Conservation Pool	Maximum Conservation Pool
	Min	Critical Fish Flow	Normal Evacuation Rate	Max Evacuation Rate	Increase per hour	Decrease per hour	Minimum Flow	Max when augmenting for mainstem flows	Increase	Decrease					
		below these rates Biop down-							**Increase per hour**	**Night-time Decrease per hour**	**Day-time Decrease per hour**	**Daily Decrease (max decrease in a 24 hour period)**			
Hills Creek	400 cfs	1500 cfs	6000 cfs	8000 cfs	100-1000cfs : 300cfs 1000-5000cfs : 500cfs 5000-8000cfs : 800cfs Maximum : 1500cfs	4000 per half hour (recommend 20%)	400 cfs		200 cfs	From 400 to 1700 cfs, 100 cfs/hour. Outflow above 1700 cfs is reduced 0.1 ft/hour at HCRO	0.2 ft/hour at HCRO	WCM - 1.5 ft/day Biop - lesser of 1 ft/day at HCRO or 50% flow volume	1448.0	1541.0
Dexter	1200 cfs	3000 cfs	12000 cfs	15000 cfs	500-1000cfs : 500cfs 1000-4000cfs : 1000cfs 4000-15000cfs:1500cfs	500-2000cfs : 700cfs 2000-5000cfs : 1500cfs 5000-10000cfs: 2500cfs 10-20 kcfs : 5000cfs	1200 cfs		0.2 ft / hour and 0.5 ft/day at DEXO	887/901	0.2 ft/hour at DEXO	WCM - 0.5 ft/day Biop - lesser of 1 ft/day at DEXO or 50% flow volume	LOP: 825.0 DEX: 690.0	LOP: 926.0 DEX:695.0
Fall Creek	50 cfs	700 cfs	3800 cfs	4500 cfs	50-1000cfs : 300cfs 1000-4000cfs : 500cfs Maximum : 800 cfs	No Limit (recommend 20%)	Feb - Mar: 50 cfs April -Aug: 80 cfs Sep 1-Oct 15: 200 cfs Oct 16-Jan 30: 50 cfs		200 cfs	From 50 to 500 cfs, 50 cfs/hour. Outflow above 500 cfs is reduced 0.1 ft/hour at FALO	From 50 to 300 cfs, 50 cfs/hour. Outflow above 300 cfs is reduced 0.2 ft/hour at FALO	lesser of 1 ft/day at FALO or 50% flow volume	728 (690 for fish)	830.0
Cottage Grove	50 cfs	800 cfs	2500 cfs	3000 cfs	350 cfs	No Limit (recommend 20%)	Feb-Jun: 75 cfs Jul-Oct: 50 cfs Nov-Jan: Inflow		100 cfs per hour 300 cfs max per day	From 50 to 800 cfs, 50 cfs/hour. Outflow above 800 cfs is reduced 0.1 ft/hour at COTO	From 50 to 300 cfs, 50 cfs/hour. Outflow above 300 cfs is reduced 0.2 ft/hour at COTO	lesser of 1 ft/day at COTO or 50% flow volume	750.0	790.0
Dorena	100 cfs	1000 cfs	4000 cfs	5000 cfs	100-2000cfs : 500cfs 2000-5000cfs : 750cfs Max per 2hrs : 1500cfs	No Limit (recommend 20%)	Feb-Jun: 190 cfs Jul-Oct: 100 cfs Nov-Jan: Inflow		200 cfs per hour 500 cfs max per day	From 100 to 1000 cfs, 100 cfs/hour. Outflow above 1000 cfs is reduced 0.1 ft/hour at DORO	From 100 to 1000 cfs, 100 cfs/hour. Outflow above 1000 cfs is reduced 0.2 ft/hour at DORO	lesser of 1 ft/day at DORO or 50% flow volume	770.5	832.0
Fern Ridge (see comment)	50 cfs	300 cfs	MNRO = 4650 cfs	3000 cfs	Normal : 750 cfs Maximum : 1000 cfs	No Limit (recommend 20%)	Feb-Jun: 50cfs Jul-Nov: 30cfs (at MNRO) Dec-Jan: Inflow		200 cfs	From 30 to 300 cfs, 40 cfs/hour. Outflow above 300 cfs is reduced 0.1 ft/hour at FRNO	From 30 to 80 cfs, 40 cfs/hour. Outflow above 300 cfs is reduced 0.2 ft/hour at FRNO	lesser of 1 ft/day at FRNO or 50% flow volume	353.0	373.5
Cougar	300 cfs	1200 cfs	5000 cfs	6500 cfs	100-500cfs : 250cfs 500-6500cfs : 500cfs Maximum : 750cfs	500 cfs	July 1 - May 30: 300 cfs June 1-30: 400 cfs		200 cfs	From 300 to 1200 cfs, 150 cfs/hour. Outflow above 2400 cfs is reduced 0.1 ft/hour at CGRO	0.2 ft/hour at CGRO	lesser of 1 ft/day at CGRO or 50% flow volume	1532.0	1690.0
Blue River	50 cfs	700 cfs	3000 cfs	3700 cfs	50 - 100cfs : 50 cfs 100-500cfs : 100cfs 500-1000cfs : 200cfs 1000-2000cfs : 400cfs 2000-3700cfs : 600cfs	30% (recommend 20%)	50 cfs		50 - 100cfs : 50 cfs 100-500cfs : 100 cfs 500-1000cfs : 200 cfs 1000-2000cfs : 400 cfs 2000-3700cfs : 600 cfs	From 50 to 700 cfs, 100 cfs/hour. Outflow above 2300 cfs is reduced 0.1 ft/hour at BLUO	From 50 to 500 cfs, 100 cfs/hour. Outflow above 500 cfs is reduced 0.2 ft/hour at BLUO	lesser of 1 ft/day at BLUO or 50% flow volume	1180.0	1350.0
Foster	Inflow up to 10000 cfs	2000 cfs	12000 - 15000 cfs	18000 cfs	500-1000cfs : 500cfs 1000-3000cfs : 1000cfs 3000-18000cfs : 1500cfs Maximum : 2500cfs	30% (recommend 20%)	Feb 1-Mar 15: 800 cfs Mar 16-May 15: 1500 cfs May 16-Jun 30: 1100 cfs Jul 1-Aug 31: 800 cfs Sep 1-Oct 15: 1500 cfs Oct 16-Jan 31: 1100 cfs	Mar 16 - May 15 3000 cfs	300 cfs	0.1 ft/hour at SSFO	0.2 ft/hour at SSFO	lesser of 1 ft/day at SSFO or 50% flow volume	GPR: 922.0 FOS: 613.0	GPR: 1010.0 FOS: 637.0
Big Cliff	1000 cfs	2000 cfs	10000 cfs	17000 cfs	100-1000cfs : 500cfs 1000-3000cfs : 1000cfs 3000-17000cfs:1500cfs Maximum : 2000cfs	No Limit (recommend 20%)	Feb 1-Mar 15: 1000 cfs Mar 16-May 31: 1500 cfs Jun 1 -Jul 15: 1200 cfs Jul 16 - Aug - 31: 1000 cfs Sep 1-Oct 15: 1500 cfs Oct 16-Jan 31: 1200 cfs	Mar 16 - May 31 3000 cfs	0.3 ft / hour and 0.5 ft/day at BCLO	0.1 ft/hour at BCLO	0.2 ft/hour at BCLO	WCM - 0.5 ft/day Biop - lesser of 1 ft/day at BCLO or 50% flow volume	DET: 1450.0 BCL 1182.0	DET: 1563.5 BCL: 1182.0

9 Total Dissolved Gas Information

The following information describes the relationship between flow and Total Dissolved Gas (TDG) at the Willamette projects. If we have to spill in order to fulfill a necessary project function, or provide flood damage reduction, or pass water in excess of powerhouse capacity, the Corps considers this involuntary spill and not a violation of the Clean Water Act and does not require a variance.

Big Cliff:
- Spillway releases above 1,500 cfs/gate will exceed the gas standard.

Blue River:
- It's estimated that a RO release above 1,500 cfs will exceed the gas standard.

Cougar:
- It's estimated that a RO release above 600 cfs will exceed the gas standard.

Detroit:
- Spillway release above 1,100 cfs/gate will exceed the gas standard.

Dexter:
- TDG information in tabular form:

	Spillway Flow by Total Dissolved Gas Percent		
Gates Open	**110%**	**115%**	**120%**
1	600	1,000	2,000
2	1,200	1,800	2,100
3	1,600	3,100	5,700
4	1,800	4,200	7,600
5	5,500	5,500	5,700
6	5,800	6,500	5,800
7	4,800	7,700	8,000

Fall Creek:
- It's estimated that RO releases above 1,500 cfs will exceed the gas standard.

Foster:
- It's estimated that SW releases above 8,800 cfs will exceed the gas standard. Dilution of spill with generating water reduces the gas downstream but it is still above the standard 5.6 km downstream.

Green Peter:
- It's estimated that RO releases above 2,000 cfs will exceed the gas standard.

Hills Creek:
- It's estimated that RO releases above 4,000 cfs will exceed the gas standard.

Lookout Point:
- It's estimated that a SW releases above 6,500 cfs will exceed the gas standard. RO releases above 2,000 cfs will exceed the gas standard.

TDG key concepts, taken from Colt 1984 and RCC / USGS common practices:

Dissolved gas is measured as a pressure: total partial pressure of dissolved gases, water, unfiltered, millimeters of mercury. Simultaneously, the barometric pressure (BP mmgHg) is measured. The barometric pressure is the indication of the equilibrium concentration of dissolved gas at the water surface. The common reporting metric is the percent of saturation of TDG:

$$TDG\ (\%\ of\ saturation) = \frac{TDG\ (mmHg)}{BP\ (mmHg)} * 100$$

This TDG (%) value used for criteria comparison and RCC reporting. Since the reported TDG (%) is calculated with barometric pressure, it represents the percent of saturation at the surface of the water, NOT at the depth of the sensor.

However, the reported TDG (%) this is not the actual TDG (%) that a fish experiences when it below the surface. The solubility of dissolved gases increase with depth because of hydrostatic pressure. Therefore, the TDG (%) decreases with depth. The actual TDG (%) that an animal experiences at depth is called the uncompensated TDG (%) (Colt 1984). Since the gauges are deeper than the surface, a fish swimming next to a gauge will experience a lower TDG (%) than the reported TDG (%). This phenomena commonly referred to as “depth compensation”. The rule of thumb is:

The TDG (% saturation) decreases by 10% for every 1 meter of depth or by 3% for every 1 foot of depth.

The solubility of dissolved gases also increase with cooler water temperature. So, as the water warms, so will the TDG (%) will increase. Another rule of thumb (Colt, 1984 Table 44):

The TDG (%) will increase by 2% for every 1°C of warming of the water.

Colt, John, 1984, Computation of Dissolved Gas Concentrations in Water as Functions of Temperature, Salinity, and Pressure.

10 Data Sources and Information

Daily information is gathered from many sources. Below is a list of just some sources of data:

- The Portland District Reservoir Regulation & Water Quality Data webpage contain a number of reports for the Willamette including daily and hourly reports. These links also provide users with additional information including graphics for the projects. The Portland District Reservoir Regulation & Water Quality Data webpage is the best site to access regulation information.

http://nwp-wmlocal2.nwp.usace.army.mil/nwp/wm_page/www/index.html

The Northwestern Division RCC also has a page with links to many sources of information.

http://nwp-wmlocal2.nwp.usace.army.mil/common/dashboard/www/index.html

All reports created by WCDS are located in various bins

http://nwp-wmlocal2.nwp.usace.army.mil/nwdp/

- That these files are only updated once per hour, usually between the 30 and 45 minute mark.
- Real-time radio gage data for the reservoirs and stream gages are available for viewing with XConnect DataView. This requires installation of DataView and setup of database connections. The Hydromet stations are shown in Figure 26. Portland District along with the operators at Lookout Point, Foster and Detroit can poll the gages through XConnect. The data is transmitted via radio to Portland District.
- Project operators can be called to get instantaneous data from gages in an emergency.
- The gages can be called through a phone line to get the latest stage reading. Phone numbers of gages are shown in Table 13.
- The "Willamette Reservoir Statistics" workbook contains several worksheets with information on project minimum and maximum flows and ramping rates.
    - \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Willamette_Basin\Regulator_Information
- Project flows, elevations, storage and percent of rule curve are summarized at a couple different links.
    - http://nwp-wmlocal2.nwp.usace.army.mil/nwdp/percentfull/www/percentfull.txt
    - http://www.nwd-wc.usace.army.mil/nwp/basins.html
- Willamette Teacup Diagram provides quick visual access to reservoirs data and current gage readings.
    - Internal, which includes forecast information
    - http://nwp-wmlocal2.nwp.usace.army.mil/nwdp/nwp_teacup/www/willamette/
    - External, which does not include forecast information
    - http://www.nwp.usace.army.mil/Missions/Water.aspx

    The "Overview" button on the left side of the map opens a pdf file with 8-day observed and 9-day forecast plots of each reservoir and control point

The "Annual" button on the left side of the map opens a pdf file with 365 days of observed data for each reservoir and control point.

- Historic and real-time stream data is available from the USGS website. Temperature and other water quality parameters are also available on the site for gages with sensors.
  - http://waterdata.usgs.gov/nwis
- The DataQuery and DBQuery websites provide access to project and stream information including multiple data streams that may not be available from some other sources. Water quality data is also available.
  - Internal - http://nwp-wmlocal2.nwp.usace.army.mil/common/dbquery/www/index.html
  - External - http://nwp-wmlocal2.nwp.usace.army.mil/nwdp/dataquery/webexec/dataquery.pl
- The National Weather Service (Portland) website provides observed and forecast weather information along with current watches and warnings. They are the official entity that issues public forecasts and flood alerts. For Willamette forecasts and flood information, the public is urged to call the NWS Portland WFO.
  - http://www.wrh.noaa.gov/pqr/
- Precipitation, temperature and streamflow forecasts are available from the Northwest River Forecast Center.
  - http://www.nwrfc.noaa.gov
  - 3-Day Weather Forecast – http://www.nwrfc.noaa.gov/weather/3_day.cgi
  - 10-Day Weather Forecast – http://www.nwrfc.noaa.gov/weather/10_day.cgi
- The most current short-term forecast (CHPS model run) is available on the teacup graphs.
- The National Weather Service Advanced Hydrologic Prediction Service website provides information for gage sites including flood stage, current flow, official forecast stream flow, and damages associated with river stage at specific sites. The official forecast is available on this website.
  - http://water.weather.gov/ahps2/index.php?wfo=pqr
- The Natural Resources Conservation Service (NRCS) provides observed and historic snow information and the seasonal water supply forecasts.
  - http://www.or.nrcs.usda.gov/snow/data/current.html
- The latest Willamette schedules are available as links on the Portland District Reservoir Regulation & Water Quality Data webpage. The schedules are also stored in a folder under the CENWP-EC-HR Outlook mailbox.
  - http://nwp-wmlocal2.nwp.usace.army.mil/nwp/schedules/www/index.html
- The latest after-hours duty list is available as a link on the Portland District Reservoir Regulation & Water Quality Data webpage. The NWP duty schedule is also stored in a folder under the CENWP-EC-HR Outlook mailbox.
  - http://nwp-wmlocal2.nwp.usace.army.mil/nwp/schedules/www/Portland_District_Weekend_Duty_List
- We update the RWCDS Dashboard with the month's projections for on-call duty.

  - http://nwp-wmlocal2.nwp.usace.army.mil/common/dashboard/www/index.html
- The Columbia River Harbor Forecast (DWOPPER) incorporates river forecasts, Bonneville outflows and Astoria tide tables to forecast harbor stages at Portland and Vancouver. This and other Columbia River information can be found on the NWD-RCC home page.
  - http://nwp-wmlocal2.nwp.usace.army.mil/common/dashboard/www/index.html
- The Emergency Action Plans describe the actions to be taken in the event of a dam safety emergency (notification procedures, emergency reservoir drawdown procedures, etc.). Hard copies are located in the Dame Safety (EC-HC) section and electronic versions are available on the Dam Safety intranet site.

  \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HC\Reference_Material\nwp-ec-gc.htm

Table 13: Phone numbers for Willamette gages.

Station ID	Station Description	Phone Number	USGS ID
ALBO	Willamette River at Albany	1-541-967-7812	14174000
BCKO	N Santiam below Boulder Creek	1-503-854-3319	14178000
BCLO	N Santiam River at Niagara	1-503-854-3130	14181500
BLU	Blue River Reservoir	1-541-822-8450	14162100
BLUO	Blue River below Reservoir	1-541-822-8324	14162200
BRBO	Breitenbush R above French Ck	1-503-854-3318	14179000
BRTO	Blue River above Tidbits Creek	1-541-822-8464	14161100
CGR	Cougar Reservoir	1-541-822-3722	14159400
CGRO	S Fork McKenzie near Rainbow	1-541-822-3168	14159500
CGWO	Cougar Weather	1-541-822-3487	NWS 351914
COT	Cottage Grove Reservoir	1-541-942-4251	14153000
COTO	Cottage Grove Outflow	1-541-942-9751	14153500
DEXO	M Fk Willamette R below Dexter	1-541-937-2362	14150000
DLLO	Tualatin River near Dilley	1-503-357-6682	14203500
DOR	Dorena Reservoir	1-541-942-3595	14155000
DORO	Dorena Reservoir Outflow	1-541-767-9957	14155500
FAL	Fall Creek Reservoir	1-541-937-3306	14150900
FALO	Fall Creek Reservoir Outflow	1-541-937-3260	14151000
FRMO	Tualatin River at Farmington	1-503-628-5684	14206500
FRN	Fern Ridge Reservoir	1-541-688-9622	14168000
FRNO	Long Tom River near Alvadore	1-541-688-9621	14169000
HARO	Willamette River at Harrisburg	1-541-995-9464	14166000
HCRO	M Fk Willamette above Salt Creek	1-541-782-2052	14145500
JASO	M Fk Willamette River at Jasper	1-541-726-9483	14152000

Table 13: Phone numbers for Willamette gages (continued).

Station ID	Station Description	Phone Number	USGS ID
JFFO	Santiam River at Jefferson	1-541-327-2291	14189000
LOCO	Lookout Creek above Blue River	1-541-822-8494	14161500
LONO	Coast Fk Willamette R at London	1-541-942-2811	14152500
LSMO	Little N Santiam R near Mehama	1-503-859-3846	14182500
MCMO	S Yamhill River at McMinnville	1-503-472-1319	14194150
MEHO	N Santiam River at Mehama	1-503-859-2641	14183000
MNRO	Long Tom River at Monroe	1-541-847-5491	14170000
MRPO	Mary's River near Philomath	1-541-929-6778	
MS	Mount Scott Repeater	1-503-808-5467	
NOTO	Long Tom River near Noti	1-541-935-1954	14166500
ORCO	Willamette River below Falls	1-503-655-7819	14207770
PP	Prairie Peak Repeater	1-503-808-5436	
SFCO	S Fk McKenzie above Cougar	1-541-822-3758	14159200
SLMO	Willamette River at Salem	1-503-581-6434	14191000
SSCO	S Santiam River below Cascadia	1-541-367-1380	14185000
SSFO	South Santiam River near Foster	1-541-367-1322	14187200
SUVO	Luckiamute River near Suver	1-503-838-2621	14190500
VIDO	McKenzie River near Vida	1-541-822-3410	14162500
WCLO	Winberry Creek near Lowell	1-541-937-9815	14150800
WNFO	Wiley Creek near Foster	1-541-367-1347	14187000
WTLO	S Santiam River at Waterloo	1-541-258-2953	14187500

10.1 Directories with Regulation Files

- The Willamette reservoir regulation files are stored here: \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\RES_REG\WILLAMETTE
- Files related to the Willamette Biological Opinion are here: \\nwd\nwp\ETDS\Willamette\ESA_Consultation.
- The specific flow management related sections are also stored here: \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\RES_REG\WILLAMETTE\AUTHORITY\BiOp_2008

10.2 Reference Documents

- National Marine Fisheries Service. 2008. Endangered Species Act Section 7(a)(2) Consultation, Biological Opinion and Magnuson-Stevens Fishery Conservation and Management Act Essential Fish Habitat Consultation, Consultation on the "Willamette River Basin Flood Control Project." Log Number F/NWR/2000/02117. National Marine Fisheries Service Northwest Region, Seattle, WA.

- Water Control Manuals \\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\RES_REG\WILLAMETTE\WCM

11 Regulation Tools

The regulators have several tools available for use to regulate the Willamette projects. These include the NWRFC CHPS model as well as regulator created spreadsheets.

11.1 Community Hydrologic Prediction Service (CHPS) Model

The NWRFC has transitioned from the NWS River Forecast System (NWSRFS) to the Community Hydrologic Prediction System (CHPS). The CHPS model is used by the regulators to evaluate the project inflow and streamflow forecasts and inform project operational decisions.

The model of the Willamette forecasts flow down to Portland on a 6-hour time step. Regulators input project regulations into the model based on forecast inflows. The model can be run more than once per day if regulation is changing. Regulators are also set up to the run the forecast from home. Below are generalized steps for running the model.

1. Log on to the Portland District server using NoMachine (remote-control software that allows you to remotely interact with the server at NWRFC).
    a. The server address is 204.228.170.158
    b. Passwords are supplied by the NWRFC (Mark Pattee).
2. Start "CHPS IFD" from the menu and follow the menu prompts to start synchronization.
3. Once you have been prompted that synchronization is complete, select "Reg Willamette" from the forecast list.
4. Select the "Modifiers" tab to update the project regulation (or mods).
5. The forecast and mods can be viewed graphically on the "Plots" tab or in tabular form.
6. Once you are satisfied with the mods, highlight the "Reg Willamette" folder and select the "Run Approved Forecast" icon. A check box will appear next to "Reg Willamette" once the forecast is done running.
7. To exit the system, select "Log Out" then "Exit" under the "File" menu.

A user's manual on the Interactive Forecast Display (IFD) component of CHPS is stored on the LAN

\\nwd\nwp\ETDS\Engineering_Division\CENWP-EC-H\CENWP-EC-HR\Reservoir_Regulation\Community Hydologic Prediction System User Manual

11.2 Spreadsheet Tools

Several spreadsheet tools have been developed by the regulators to assist with Willamette regulation. These may be updated periodically with new information or to make them more user-friendly. Below is a description of the spreadsheets currently available:

- **Will_Schedules_20XX.xlsm** – Spreadsheet developed to prepare the Willamette non-power, baseload and power schedules. This is described in more detail in Section 2.2.1.
- **CWMS Daily Update for 20XX.xlsm (filed under the sub-folder Graphs)** – Spreadsheet tool developed to summarize daily project operations, help forecast seasonal inflows and outflows, develop the summer augmentation plan and graphs, check amount of augmented flow needed

in the summer months, forecast future pool levels and to develop AER. This spreadsheet is described in more detail in Section 11.3.

- **Interim draft limits and Interim draft get data (filed under the sub-folder Graphs)** – Spreadsheet used calculate the required storage in the reservoirs (interim draft limit) to meet the BiOp flow targets through October. The method was developed after the drought of 2001 for estimating interim draft limits. The current method assumes an empty pool on October 31st and 25% of the period of record average inflow. The outflows required by the BiOp are applied and the daily required storage to meet those flows is back-calculated to mid-May. If the inflow is below 25% of average the pool will draft more quickly. The method also assumed that after October, the projects would be passing inflow. Users can look at different levels of inflow and outflow to see the results for each reservoir graphically. Observed data should be copied from the "Data_Page" worksheet in the "Daily_Status_of_Portland_Reservoirs.xls" workbook to the "Obs" worksheet in the "Interim draft limits 2012.xls" workbook.
- **New Willamette Crystal Ball.xlsx (filed under the sub-folder Graphs)** – a new spreadsheet that was developed during the droughts of 2015 and 2016 to better identify how deep we would draft the project with select inflow/outflow scenarios.
- **daily project graphs-20XX.xlsm** – This spreadsheet graphs the rule curve, estimated elevations, inflow and outflow for the year as calculated in the Augmentation spreadsheet (see Section 11.3.3). The graphs in this workbook are used in the augmentation plan. The graphs can be updated by copying the "summerblk" worksheet from the "CWMS Daily Update 20XX.xlsm" workbook to the "summerblk" worksheet in this workbook. **Copy and Paste Special Values only – do not copy formulas.**

11.3 CWMS Daily Update Workbook

This spreadsheet is used summarize daily project operations, help forecast seasonal inflows and outflows, develop the summer augmentation plan and graphs, check amount of augmented flow needed in the summer months, forecast future pool levels and to develop AER. It is important that this spreadsheet be maintained since it is used for so many important tasks. All cell values need to be filled and positive values. Below is a description of some of the worksheets within the workbook. The more involved worksheets in the workbook are described in Sections 11.3.1 through 11.3.3 below.

- **Get Data** – Used to pull recent data from the CWMS database, populate the workbook and post project summary information and graphs to the web. Copy the data to the Importsheet tab. This will propagate data to the Stats and Graphs tabs. Manually interpolate if there is missing data.
- **Stats** – Contains daily summary information for the projects including storage, elevation, inflow, outflow and deviation from rule curves.
- **Graphs** – Contains graphs of all the projects showing rule curve, elevation, inflow and outflow.
- **MinAuthQ** – Contains the minimum congressionally authorized flows listed in the Water Control Manuals and the authorizing legislation (House Document No. 531 Volume V). These are different than the BiOp tributary flows. This sheet also contains the BiOp mainstem and tributary flow targets, augmentation hierarchy, and ramp rate requirements.

- **2.1DATA**– The data pulled from CWMS populates this worksheet. This data must then be copied and pasted to the "ImportSheet" worksheet. It is very important that this spreadsheet be maintained. Typically data is pulled in Monday and Thursday in preparation for briefings.
- **ImportSheet** – Recent data from the "DATA" worksheet is copied to this worksheet. This data is then used by the subsequent worksheets in the workbook.
- **Log** – This worksheet provides information about the data pulled from the database. It is populated when the "Get Data" macro is run on the "Getx.xData" worksheet.
- **Forecast** – Contains 60 year average data for the Willamette. This forecast is used for computing power pool forecast. Typically we leave the forecast at 100% avg but cell O1 can be adjusted if the forecast graphs look wrong (used to increase or decrease local flows).
- **Fill Tables** – Summarizes project information for the year. This worksheet can be used in graphs.
- **RuleCurve** – Contains the storage and elevation rule curves for each project.
- **aeracap** – Contains elevation/storage tables for each project.
    - **Help** – Contains information on the macro that pulls data from the CWMS database on the "SetupNWD" worksheet. Also refer to the WICI at

http://nwp-wmlocal2.nwp.usace.army.mil/wiki/doku.php?id=applications:cwms_excel_spreadsheet

- **Units** – Contain information on acceptable units for different parameters (elevation, area, etc.)

11.3.1 Power Pool Worksheet (pwrpool)

This worksheet serves as the template for the AER data submittal required twice a month by the Northwest Power Pool, as described in Section 2.2.3. The submittal portion is located toward the bottom of the worksheet. It can be accessed easily by selected the named cell "PPLETTER". The upper portion of the worksheet calculates the necessary information.

Two types of data are needed for the data submittal:

1) End of month draft limits (in ksfd of draft from full pool) can be found in rows 43 – 48 of the "pwrpool" worksheet.
2) AER streamflow updates (inflows) can be found in rows 19 – 25 of the "pwrpool" worksheet.

Below are some notes on the "pwrpool" worksheet:

- Always show the actual data from the last period. Forecast data should go through the date specified on the data submittal reminder or specified in cells B110-B121.
- When referencing the cells located in the columns above the template, be sure to check that the reference is pointing to the correct cell.
- When completing the data submittal during the first part of the month, assume 100% of the 60-year median inflow.
- For the second submittal of the month estimates have been made for flows for the last week of the current month.
- The spreadsheet assumes that the current period inflows from "summerblk" worksheet are correct and uses those estimated values. It is important to go to the current period inflows and

adjust them – use the greyed cells to adjust based on %avg flows. EX: adjust February flows because it was a dry month – go to ROW81 – see the light grey numbers, typically 1.0? Adjust these so that the inflows for the future days of the month are positive (change the 1.0 to maybe a 0.75), and in line with the current forecast (at a minimum, adjust so that they are NOT negative). This really is the only adjustment that needs to be made in the worksheet. COPY and PASTE SPECIAL VALUES AND NUMBER FOREMATS to SummerblockGraphs20XX.xlsm to ensure that the elevations look correct (past and future). Adjust as necessary to make the graphs look correct – this data will translate back to the pwrpool worksheet and those numbers will be ready to send to NWD.

- Lost Creek data is included in the CWMS lookup, so is incorporated in the submittal.

11.3.2 Summerblk Worksheet (summerblk)

This worksheet is used for longer planning of reservoir operations for the AER and for quick modeling of system or individual project operations. It provides a location to store daily inflow, outflow, elevation, and control point flow data for graphing in the summerblock graphs worksheet.

The "summerblk" worksheet is where all data is manipulated. Since the worksheet is so large, data has been color-coded to be recognized easily.

Minimum Flows (Columns G to AC, AL and BA) – yellow

Project Inflows (Columns AK to BC) – pink

Lake Storage (Columns BL to BX) – red

Control Point Flows (Columns F to AI) – light green

- **Minimum Flows** – Enter minimum flows into the yellow cells at the beginning of the periods in columns G to AC, AL and BA (BIOP FLOWS ARE ALREADY ENTERED). When the spreadsheet calculates the end of month elevation, it checks to make sure that the minimum release is maintained. If the calculated flow is below the minimum, the end of month storage is dropped to show that the project will have to draft to maintain minimum releases. If the calculated release is above minimum outflows, the value is shown in the summary tables of the Augment worksheet, "augmntblk", along with the target storage. The spreadsheet will not show the project going above the rule curve. Be sure to use minimum flows from the augmentation plan, as opposed to the minimum flows shown in the project operating limits book. Minimum flows may need to be raised if target flows are NOT being met at Salem or Albany.
- **Project Inflows** – When in the current period, the project inflows can be adjusted by multiplying the yellow block at the bottom of each column, by a factor. The last number of each section is the average of the period. When you enter in a factor (LIGHT GREY NUMBER, TYPICALLY SET TO 1.0 AT BEGINNING OF THE YEAR), the spreadsheet will average the actual data and do a straight-line calculation for the remainder of the period, so that the average for the period equals the estimation you entered at the bottom of the period. ADJUST TO ENSURE THAT NO NUMBER IS ZERO or you will get a devide by zero error in the calculations!

- **Lake Storage** – The projects' lake storage in "summerblk" worksheet are all calculated. The only way to change them is to change the target elevations in cells BM105:CF120 of the Augment worksheet, "augmntblk", or lower the minimum flows. If minimum flows cause the project to draft, it will show in project summaries in the "augmntblk" worksheet and can also be seen in the lake storage columns of the "summerblk" worksheet. This is a good place to see the effects of the summer augmentation plan.
- **Control Point Flows** – The control point flows use the calculated outflows and the forecasted locals to estimate the flows at the downstream control points. Actual data is used if it is available and the forecast data is used for all days in the future.

11.3.3 Augmentation Worksheet (augmntblk)

The Augment worksheet is basically filled with summary tables. Although it was originally created for the summer months, this worksheet is now used throughout the entire year's Power Pool submittal. Figure 27 outlines the decision tree used in the worksheet for determining outflows. Each month is broken into half months (except for June and September). There are 3 time windows used throughout the spreadsheet:

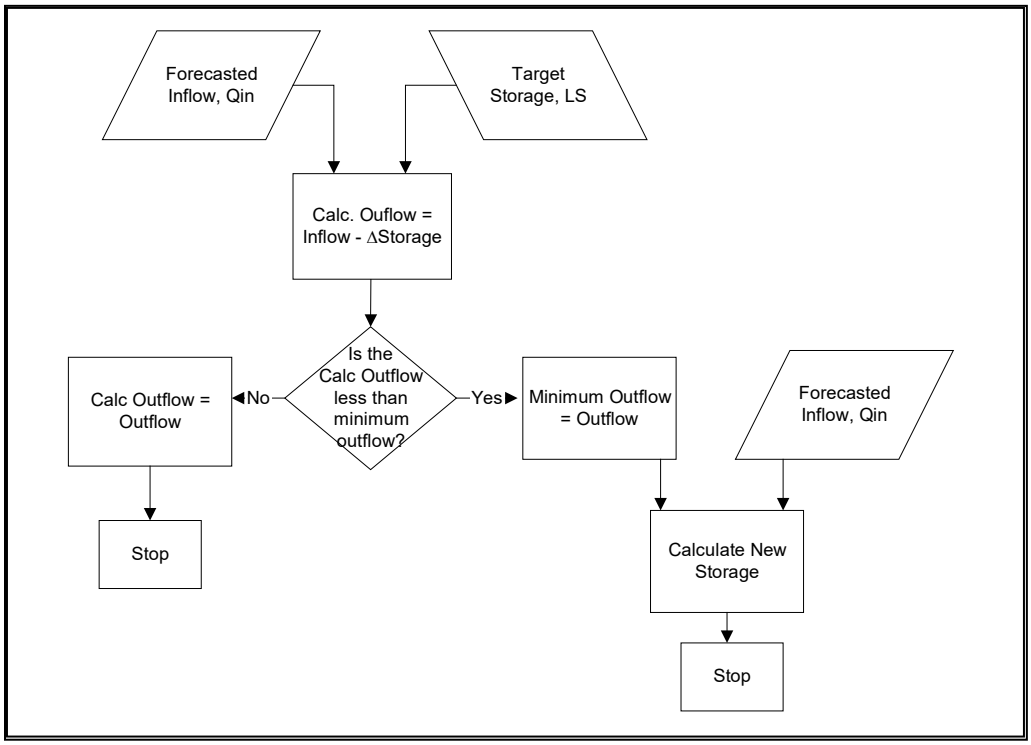
1) Actual data is used for T-1 day and beyond, where T is today.
2) Forecast data is used for period N+1, where N is the current period, and
3) A modified forecast is used when currently in period N.

The NRCS issues a water supply forecast at the beginning of each month from February through July. These are inflow volume forecasts for the projects. The forecast periods are April – September and August – September. NWP also averaged 60 years of inflow for the Willamette Basin based on Olympic smoothing (see DSSMATH Manual). The daily data was then averaged into the same periods described above (half months except for June and September). Using another spreadsheet (Unit-hydrograph-MAY.xls), the volume forecast was input and the inflows were adjusted up or down, depending on the forecast. These values are input into the augment spreadsheet as forecasted inflows.

Target storages are entered into cells BM105:CF120. Using the forecasted inflows, the necessary outflows are calculated to reach the target storages. The worksheet checks to be sure minimum releases are met. If the calculated releases are below minimum, the end of month elevation is recalculated using the minimum release.

Target flows for Albany and Salem are entered into cells BM61:CD68. Future Albany and Salem flows are forecasted using calculated releases and NRCS forecasted local flows. If the target flows are not going to be met, the additional amount needed is displayed in the period summaries. These values should be noted for future planning. If it looks like mainstem target flows are not going to be met, minimum flows should be raised in the "summerblk" worksheet until targets are met.

Figure 16: Decision tree used in the "augmntblk" worksheet to determine outflows.



11.4 Sample Calculations

- Hourly Inflow Calculation
    1. Determine change in storage in acre-feet/day.

       $LakeStorage_{current} - LakeStorage_{1\ hour\ ago} = \Delta Storage(acre\text{-}feet/hour)$

       If $\Delta Storage$ is positive, project is filling.

       If $\Delta Storage$ is negative, project is drafting.

    2. Convert ΔStorage from acre-feet/hour to ft^3/sec.

       $1\ ft^3/sec = 1.9835\ acre\text{-}feet/day = 0.0826\ acre\text{-}feet/hour$

       $\Delta Storage(acre\text{-}feet/hour) / (0.0826\ acre\text{-}feet/hour) = \Delta Storage(ft^3/sec)$

    3. Compute hourly inflow.

       $Inflow\ (ft^3/sec) = \Delta Storage\ (ft^3/sec) + Outflow\ (ft^3/sec)$

- Fill Rate (or Draft Rate) Calculation
    1. Determine the storage difference from rule curve in acre-feet.

       $LakeStorage_{current} - LakeStorage_{desired} = \Delta Storage(acre\text{-}feet)$

    2. Convert storage from acre-feet/day to ft^3/sec.

       $1\ ft^3/sec = 1.9835\ acre\text{-}feet/day$

       $\Delta Storage(acre\text{-}feet/day) / (1.9835) = \Delta Storage(ft^3/sec)$

    3. Calculate the rate of fill (or draft) in $ft^3/sec/day$ to reach desired storage in a specified number of days.

       $Rate\ (ft^3/sec/day) = \Delta Storage(ft^3/sec) / Number\ of\ Days$

Eugene Water Right: The right (Certificate 59549) is for 2000 cfs from June 1 through October 31. It is 2500 cfs from November 1 through May 31. Based on the limited measurement we have done I think the gage is fairly accurate during the summer. No idea in the winter. Before he realized that the gage was there the prior Watermaster would just add the flow at Goshen and Jasper to estimate the flow at Eugene. That actually is an effective way of estimating what the flow is at Eugene.

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